BEYOND HULTEN: AI AND THE OPEN-ECONOMY INPUT-OUTPUT DECOMPOSITION*

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October 2026

Most recent draft here.

This paper quantifies the aggregate productivity consequences of artificial intelligence in a 77-economy general equilibrium model calibrated to OECD Inter-Country Input-Output tables. Extending Baqaee and Farhi (2019) to an open economy with Armington (1969) preferences and a task-based AI microfoundation, we derive three formal propositions and a five-component welfare decomposition. Under simultaneous global adoption, advanced economies (AEs) gain 2.35 percentage points per year (pp/yr) in labor productivity—2.5 times the 0.93 pp/yr closed-economy baseline—while emerging markets (EMs) and developing economies (DEs) gain 1.33 pp/yr and 0.85 pp/yr, respectively. The 1.02 pp/yr AE-EM gap holds in 97.2 percent of 1,000 Monte Carlo draws. The international input-output spillover dominates the terms-of-trade penalty by 1.3–4.2 times across all 77 economies, guaranteeing a positive net open-economy gain everywhere. Digital trade liberalization adds 0.122 pp/yr for AEs and 0.055 pp/yr for emerging economies at zero additional technology cost. The closed-economy benchmark is not a conservative lower bound but a structurally misleading baseline.

JEL Codes: O33, F41, D57, O47, F15, D24, O11.

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I. INTRODUCTION

AI-enabling goods exports grew from $1.0 trillion to $3.8 trillion between 2014 and 2025 (WTO 2025), and AI-related services trade tripled to $2.1 trillion over the same period (UNCTAD 2023)—structural evidence that AI's productive capacity crosses borders at near-zero marginal cost and that its macroeconomic consequences cannot be adequately analyzed in a closed-economy framework. Unlike earlier automation technologies that were largely confined to routine physical tasks, AI processes language, images, and complex decisions across finance, legal services, software development, and data analytics—sectors whose output is geographically concentrated and directly tradable, making AI's diffusion inseparable from global value chain architecture (Figure 1).[1] A closed-economy model of AI productivity is therefore not a conservative lower bound but a structurally misleading baseline that misattributes the source, scale, and cross-country distribution of welfare gains.

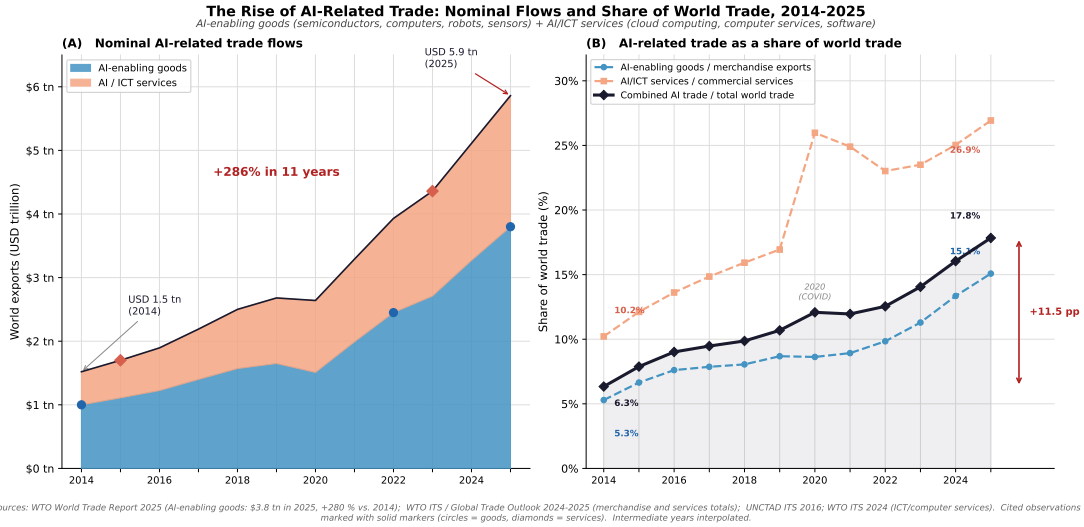


Figure 1. Structural transformation of global AI-related trade, 2014–2025. Left panel: AI-enabling goods exports ($1.0T to $3.8T, 2014–2025). Right panel: ICT services exports ($0.58T to $2.1T, 2014–2025). Source: (WTO 2025), (UNCTAD 2023).

This paper extends the Baqaee and Farhi (2019) input-output framework—itself a nonlinear generalization of Hulten's (1978) theorem—to an open economy with Armington (1969) constant-elasticity preferences over differentiated national varieties. The model is calibrated to the OECD Inter-Country Input-Output tables (OECD 2023), spanning 77 economies and 45 ISIC Revision 4 sectors and yielding a $3{,}465 \times 3{,}465$ world Leontief inverse through which productivity shocks propagate across national borders. AI's effect on

[1] ICT services exports represent the primary vector for AI capability diffusion through cloud computing and software.

sectoral TFP is microfounded in the Acemoglu and Restrepo (2018) task-augmentation framework, with sector-level AI exposure scores from Eloundou et al. (2023) and Felten et al. (2021) and a per-task productivity gain calibrated from firm-level evidence (Brynjolfsson et al. 2023; Dell'Acqua et al. 2023). The US closed-economy benchmark of 0.93 pp/yr under broad adoption exactly matches Filippucci et al. (2024), providing external validation before any open-economy results are presented.

The central theoretical contribution is three formal propositions characterizing the mechanics and distribution of AI gains in an open economy. Proposition 1 derives the open-economy Hulten formula: each country's aggregate TFP gain equals the inner product of trade-weighted Domar weights (rows of the world Leontief inverse normalized by GDP) and the world AI productivity shock vector, nesting the Baqaee and Farhi (2019) closed-economy result as a special case. This result shows how global supply-chain linkages transmit foreign AI gains to domestic welfare. Proposition 2 characterizes the terms-of-trade deterioration facing AI exporters and proves that the international IO spillover strictly dominates it by a factor of 1.3–4.2 across all 77 economies—so the net open-economy gain is unambiguously positive. Proposition 3 establishes the income-group divergence condition and shows how digital trade liberalization compresses the AE-EM gap. The three propositions are unified by a five-component welfare decomposition that nests the closed-economy three-component result as the special case when trade is shut down.

Under simultaneous adoption across all 77 economies at country-specific rates, AEs average 2.35 percentage points per year (pp/yr) in labor productivity growth against 0.93 pp/yr in autarky—a factor of 2.5 attributable to global supply-chain transmission—while EMs average 1.33 pp/yr and DEs 0.85 pp/yr.[2] For the US specifically, the open-economy gain is 1.87 pp/yr, exactly twice the closed-economy baseline, with approximately 50 percent originating abroad through imported intermediate price reductions. Small, highly trade-integrated economies rank highest: Hong Kong (5.28 pp/yr), Luxembourg (3.27 pp/yr), Singapore (3.21 pp/yr), and Ireland (3.18 pp/yr). The AE-EM productivity gap of 1.02 pp/yr is robust in 97.2 percent of 1,000 Monte Carlo draws, decomposing into three gap-widening forces—adoption-rate differentials, domestic supply-chain density, and GVC centrality—partially offset by terms-of-trade and Baumol penalties that fall more heavily on AEs.

The scenario analysis delivers four additional results. Under US-only adoption, Ireland (0.38 pp/yr), Canada, and Mexico receive positive spillovers through GVC linkages despite zero domestic adoption, and GVC integration ($R^2 = 0.71$) is the dominant predictor of spillover receipts across all 77 economies—confirming that supply-chain position, not income level, deter-

[2]These differences in AI-related productivity gains across income groups are consistent with the findings of a recent IMF departmental paper focusing on Sub-Saharan Africa (Schindler et al. (2026)).

mines which countries benefit from a single large adopter. Trade fragmentation reduces AE gains to 2.20 pp/yr and EM gains to 1.28 pp/yr relative to the full-adoption baseline.[3] Digital trade liberalization for AI-intensive service sectors adds 0.122 pp/yr for AEs and 0.055 pp/yr for EMs—a 13 percent bonus for AEs achievable through trade-policy coordination alone, without additional investment in AI technology.

A concurrent and methodologically independent analysis, Filippucci et al. (2026) — hereafter FGLSY (2026), estimates 0.10–0.95 pp/yr in per-capita real income gains for OECD and G20 economies using a calibration framework with knowledge spillovers. The convergence of two independent papers on the primacy of open-economy channels provides cross-methodological corroboration for the central finding. The papers differ in scope (77 economies including 26 EMs and 10 low-income DEs versus OECD/G20), theoretical architecture (three formal propositions with explicit sufficient conditions versus a quantitative calibration framework), the isolation of the pure trade-spillover channel via Scenario T2, and welfare metric (long-run labor productivity via the Ramsey–Cass–Koopmans multiplier versus per-capita real income at the transition path). Methodologically, three contributions stand apart from the AI application itself. First, trade-weighted Domar weights derived from a calibrated 77-economy world input-output system—combined with a five-component welfare decomposition—nest the Baqaee and Farhi (2019) result as a special case while extending it across borders. Second, Scenario T2 isolates the pure spillover channel, showing that supply-chain position—not domestic adoption rate—drives the cross-country distribution of gains from a single large adopter. Third, a formal proof that the IO spillover strictly dominates terms-of-trade deterioration guarantees positive net open-economy welfare for all 77 economies. Proposition 3 extends the distributional literature (Helpman et al. 2010; Autor et al. 2013) from within-country wage effects to between-country AI productivity gaps.

Section II builds the case for an open-economy framework across five structural channels. Section III positions the paper in the literature. Section IV develops the model and the three propositions. Section V describes the calibration. Section VI presents results and Section VII concludes. Proofs appear in the Appendix.

II. WHY INTERNATIONAL TRADE IS INDISPENSABLE: FIVE CHANNELS

A closed-economy analysis of AI productivity is not a conservative lower bound—it is a structurally misspecified baseline that suppresses the dominant transmission mechanism. The case for an open-economy framework rests on five distinct channels, each operating independently and, in most

[3]This result represents a Pareto deterioration in which all income groups lose from barriers that slow supply-chain transmission.

cases, reinforcing the others: supply-chain transmission through the world input-output network, cross-border diffusion of AI as a non-rival good, terms-of-trade redistribution between adopting and non-adopting economies, comparative advantage dynamics that tilt welfare gains toward knowledge-intensive sectors, and the digital liberalization premium. The following subsections develop each channel in turn. Section IV formalizes them.

The world economy is organized through global value chains, not independent national production systems, and this structural fact makes AI a fundamentally open-economy phenomenon. Approximately 70 percent of world merchandise trade consists of intermediate goods and services traded for further processing (WTO 2019). An AI-driven productivity improvement in US software lowers the price of software intermediates that German manufacturers, Korean electronics firms, and Mexican assembly plants all source as inputs. This propagation operates through the Leontief inverse of the world input-output matrix—it is the dominant transmission mechanism, not a second-order refinement.

Two features of the ICIO data amplify this supply-chain effect considerably. First, the sectors with the highest AI exposure—Finance, Information Technology, Professional Services, Telecommunications—are also the most pervasive upstream suppliers in the OECD ICIO network. Finance alone carries a Domar weight (0.154) that is 79 percent above its value-added GDP share (0.086), meaning that a productivity gain in Finance propagates further through the economy than an equal-sized gain in any sector whose Domar weight is close to its GDP share. Second, for the median AE, imported intermediates account for 25–40 percent of total intermediate use, so domestic AI adoption alone is far from sufficient to capture the full supply-chain dividend. The calibration confirms this: the sum of US Domar weights equals 1.762, implying that every dollar of US final demand requires $1.76 of gross output—a multiplier that closed-economy analyses cannot recover.

AI models are non-rival in use: once trained, a large language model or a computer-vision system can be deployed simultaneously by any number of users at near-zero marginal cost. This distinguishes AI from physical capital (rival and geographically bound) and from human capital (geographically immobile), making AI potentially the most globally diffusible general-purpose technology in history. Non-rivalry does not make AI perfectly diffusible in practice. Intellectual property protections, export controls, data-localization regulations, and compute-access barriers all restrict the international flow of AI capability. But these barriers operate through trade channels—they impede imports of AI-enabled services and outputs, not just underlying model weights. A closed-economy model implicitly assumes these barriers are absolute and permanent, an assumption that is both empirically inaccurate and normatively unhelpful. An open-economy model with calibrated Armington barriers captures the intermediate and more realistic case.

The structural transformation of global trade documented in Figure 1

quantifies the empirical pressure toward an open-economy framework. AI-enabling goods— semiconductors, industrial robots, sensors, computers, and telecom equipment— grew from approximately $1.0 trillion in 2014 to $3.8 trillion in 2025, raising their share of world merchandise exports from 5.3 to 15.1 percent (WTO 2025). Simultaneously, ICT and computer-services exports rose from $580 billion to $2.1 trillion, increasing their share of world commercial services from 10.2 to 26.9 percent (UNCTAD 2023; WTO 2025). Supply in both categories is geographically concentrated in a small number of AEs, making closed-economy analyses systematically incomplete as a guide to the cross-country distribution of AI's productivity dividend.[4]

A third channel absent from closed-economy models is terms-of-trade deterioration. When a country's production costs in AI-exposed sectors fall, competitive pressure forces export prices downward, partially offsetting the domestic welfare gain—the export-curse mechanism identified by Bhagwati (1958). In the AI context, the mechanism operates through knowledge-intensive services: an early-adopting country that produces cheaper software, financial analytics, and professional services sees the unit value of those exports decline even as volumes rise. The mechanism also creates a free-rider incentive: a country that delays adoption while its partners adopt enjoys falling import prices for AI-enhanced goods without bearing the export price deterioration. This provides a theoretical rationale for international coordination on AI diffusion that closed-economy analyses cannot identify.

AI adoption creates comparative advantage dynamics that closed-economy models suppress entirely. The sectors with high AI exposure—Finance, Information Technology, and Professional Services—are concentrated in AEs. Sectors with low AI exposure, such as labor-intensive manufacturing and basic agriculture, are more prevalent in developing economies. AI therefore differentially amplifies the sectors AEs are already specialized in, tilting welfare outcomes toward economies that were already knowledge-intensive. Emerging economies partially compensate through trade by importing AI-enhanced services as intermediate inputs. The five-component welfare decomposition in Section IV.J formally isolates this channel and quantifies its contribution to the income-group divergence established in Proposition 3.

The fifth channel is specific to AI as a digital technology: the sectors that produce the most AI-amenable outputs—software, financial services, legal and professional advice, data analytics—are also the sectors in which barriers to trade have been falling most rapidly through cloud computing, data-interconnection agreements, and digital trade chapters in modern free-trade agreements. Digital liberalization therefore functions as a force multiplier for AI productivity gains rather than an independent policy instrument: the same service-trade opening that expands market access also accelerates the

[4]For instance, Asian economies account for approximately 65 percent of global AI-enabling goods exports, while the US accounts for approximately 87 percent of cloud-computing service exports.

cross-border transmission of AI capability improvements. Scenario T4 in Section VI.D quantifies this channel and shows that digital liberalization has an asymmetric equalizing effect on service-trade flows, contributing to the income-group convergence documented in Proposition 3(ii).

Taken together, these five channels imply that the closed-economy benchmark misses the dominant transmission mechanism, misattributes the cross-country distribution of gains, and misidentifies the policy levers available to non-frontier economies. Section IV formalizes each channel and derives the conditions under which the open-economy gain strictly dominates its closed-economy counterpart.

III. RELATED LITERATURE

The two papers closest to this one are Filippucci et al. (2024) and FGLSY (2026). Filippucci et al. (2024) apply the Baqaee and Farhi (2019) input-output framework to the United States and report that broad AI adoption raises labor productivity by 0.93 percentage points per year—the benchmark this paper uses as its primary closed-economy calibration checkpoint in Section V.D. FGLSY (2026) provide the first open-economy quantification for OECD and G20 economies, finding per-capita gains of 0.10–0.95 pp/yr with GVC spillovers of 30–40 percent for OECD subgroups. This paper extends both along three dimensions: scope (77 economies, including 26 emerging and 10 developing countries absent from prior open-economy analyses), theoretical architecture (three formal propositions with explicit sufficient conditions and a five-component welfare decomposition), and isolation of a pure trade-spillover channel through a novel counterfactual scenario. The convergence of both papers on the primacy of open-economy trade channels provides cross-methodological corroboration for the central quantitative finding.

The theoretical backbone is the input-output productivity literature. Hulten (1978) showed that a sector's contribution to aggregate TFP equals its Domar weight—its gross-output share of GDP—a sufficient-statistic result that holds in competitive equilibria. Baqaee and Farhi (2019) generalize the Hulten theorem to non-competitive economies, deriving a second-order correction that captures reallocation frictions and the Baumol (1967) drag when gains concentrate in fast-growing, price-falling sectors. Carvalho and Tahbaz-Salehi (2019) and Acemoglu et al. (2012) establish that network topology and sectoral heterogeneity shape propagation nonlinearly: hub sectors amplify shocks, and the sign of second-order effects depends on the curvature of the production network. This paper extends the Baqaee–Farhi framework from a closed economy to a multi-country system, replacing the domestic Leontief inverse with a world Leontief inverse that spans all bilateral trade flows. The open-economy Hulten theorem (Proposition 1) nests the closed-economy result as the special case in which all bilateral trade

shares collapse to the identity matrix.

The scenario range in Section V draws on contrasting positions in the AI productivity literature. Brynjolfsson et al. (2021) document a productivity J-curve: adoption of general-purpose technologies first displaces complementary organizational capital, generating a temporary output shortfall before long-run gains materialize, a pattern that rationalizes pessimistic near-term and optimistic long-run forecasts simultaneously. Acemoglu (2024) develops a more cautionary view, arguing that aggregate productivity gains may be limited if automation substitutes for labor in low-value tasks or crowds out the complementary human investment that generates sustained productivity growth. Against these cautions, firm-level evidence from Brynjolfsson et al. (2025) and Dell'Acqua et al. (2026) documents large within-firm productivity gains from AI tools in service occupations, anchoring the upper end of the per-task improvement parameter in calibration. I read these divergent estimates not as a bounded sensitivity interval around a consensus but as genuine uncertainty about the magnitude of AI's aggregate effect.

The model's AI exposure function is microfounded in the task framework of Acemoglu and Restrepo (2018), who model labor-displacing automation alongside new task creation within a production structure that maps aggregate employment and productivity effects to the task content of work. Sectoral AI exposure scores are constructed by applying the occupation-level exposure mappings of Eloundou et al. (2023) and Felten et al. (2021) to the occupation-task taxonomy developed by Acemoglu and Restrepo (2022) and Autor et al. (2003). Country adoption rates are calibrated from McKinsey (2023) and OECD (2023b), with the adoption-lag heterogeneity documented by Comin and Hobijn (2010)—up to 50 years for major general-purpose technologies—motivating the income-group multipliers applied to emerging and developing economies. The task-offshoring channel of Grossman and Rossi-Hansberg (2008) is conceptually adjacent but operationally distinct: it models the fragmentation of individual production stages across borders, whereas this paper's open-economy channel operates through intermediate-good commodity flows in the world Leontief system.

The trade block is built on the Armington (1969) constant-elasticity-of-substitution demand structure, with the elasticity of substitution calibrated from Caliendo and Parro (2015). Good-level heterogeneity from Broda and Weinstein (2006) informs the sensitivity analysis in Section VI.E, and Eaton and Kortum (2002) and Hertel (1997) provide the general-equilibrium welfare framework that maps trade flows to productivity and income. The empirical backbone is the OECD Inter-Country Input-Output tables (OECD 2023), covering 77 economies and 45 sectors in reference year 2020, which yield the $3{,}465 \times 3{,}465$ world Leontief inverse. The structural transformation of global trade documented in Figure 1—AI-enabling goods exports growing from $1.0 to $3.8 trillion between 2014 and 2025 (WTO 2025), ICT-services exports tripling to $2.1 trillion (UNCTAD 2023), with supply geographi-

cally concentrated in a small number of advanced economies—validates the open-economy approach and makes closed-economy analyses structurally incomplete rather than merely conservative.

Two empirical regularities in the technology diffusion literature motivate the model's treatment of adoption heterogeneity. Keller (2004) shows that R&D spillovers are geographically concentrated among close trading partners, with distance elasticities implying a roughly threefold intensity premium for trade-linked partners relative to arm's-length ones—a pattern that makes GVC participation a channel for AI capability transfer as well as for goods trade. Comin and Hobijn (2010) document adoption lags of up to 50 years for major general-purpose technologies and large variation across countries at comparable income levels. Both regularities motivate the income-group adoption multipliers applied in Section V: Keller (2004) by making spillovers endogenous to trade linkages, and Comin and Hobijn (2010) by embedding cross-country heterogeneity in productivity gains into the model's structure rather than treating it as a residual.

On the distributional side, the results connect to the literature on trade and inequality. Helpman et al. (2010) show that trade liberalization raises within-country wage inequality through firm-selection effects, with the sign depending on the elasticity of labor supply to the expanding export sector. Autor et al. (2013) document the regional labor-market consequences of the China trade shock, establishing a methodology for attributing distributional outcomes to specific trade exposures. This paper extends the analysis to between-country inequality in AI productivity gains—a dimension that trade exposure shapes but that neither study addresses directly—through Proposition 3. Digital trade liberalization (Scenario T4) adds a precisely characterized equalizing channel through its asymmetric impact on service-trade flows (Proposition 3(ii)), connecting the AI quantification literature to the international inequality literature in a direction not previously formalized. The direction and magnitude of both effects are established in Section VI.

I measure welfare as long-run labor productivity at the Ramsey–Cass–Koopmans steady state (Cass 1965; Koopmans 1965), rather than TFP alone. Converting a permanent TFP gain to a labor-productivity gain requires capital deepening: when AI permanently raises TFP, the steady-state capital–labor ratio rises until the net marginal product of capital is restored to the discount rate, amplifying output per worker beyond the immediate TFP gain by a factor that depends on the capital income share. Gollin (2002) and Mankiw et al. (1992) supply the cross-country factor-share estimates used to construct this multiplier. The choice of this metric, rather than per-capita real income at the transition path, is the principal welfare-metric difference between this paper and FGLSY (2026), and it accounts for a portion of the gap between reported magnitudes.

IV. THE MODEL

We build the model in four steps, each adding a new element of the production structure. Sections IV.A–IV.B define the firm problem and competitive equilibrium. Section IV.C provides the task-level microfoundation for AI's effect on TFP through the task-augmentation framework of Acemoglu and Restrepo (2018). Section IV.D derives the Hulten aggregation theorem and the Baumol second-order correction. Section IV.E opens the economy to trade using Armington (1969) preferences and derives the three propositions and five-component welfare decomposition.

IV.A. Production structure

Consider a world economy with C countries and N sectors. Let c range over $\{1,\ldots,C\}$ and i,j over $\{1,\ldots,N\}$. In each country-sector pair (c,i), a representative firm produces gross output Y_{ci} using capital, labor, and intermediate inputs purchased from all other country-sector pairs. Production follows a four-layer Cobb-Douglas nest.

In the first layer, the firm combines its productivity A_{ci} with a composite G_{ci} of value added and intermediates:

$$Y_{ci} = A_{ci} \cdot G_{ci}(V_{ci}, M_{ci}) \tag{1}$$

where $A_{ci} \geq 1$ is Hicks-neutral Total Factor Productivity (TFP).[5]

In the second layer, value-added V_{ci} is a Cobb-Douglas aggregate of capital K_{ci} and labor L_{ci}:

$$V_{ci} = K_{ci}^{\alpha_K} \cdot L_{ci}^{1-\alpha_K}, \quad \alpha_K = 1/3 \tag{2}$$

with capital share $\alpha_K = 1/3$, reflecting the empirical regularity that labor earns about two-thirds of national income across market economies (Kaldor 1961; Mankiw et al. 1992; Hall and Jones 1999; Gollin 2002; Elsby et al. 2013) and M_{ci} is a Cobb-Douglas composite of intermediate inputs sourced from all country-sector pairs:

$$M_{ci} = \prod_{c',j} \left[M_{c'j,ci}\right]^{\gamma_{ci,j} \cdot B_{c'c}} \tag{3}$$

where $M_{c'j \to ci}$ denotes the quantity of good (c',j) used as intermediate in (c,i). While Filippucci et al. (2024) ground intermediate input demand in a Leontief (fixed-coefficient) assumption, this paper nests it within a Cobb-Douglas structure. Adopting the standard Leontief fixed technical coefficients gives rise to four problems. First, Hulten's theorem breaks down under pure Leontief technology because the envelope theorem no longer applies, and the Domar-weight formula is therefore no longer valid. Second,

[5] Hicks-neutral TFP implies that, for instance, doubling A_{ci} doubles output for the same inputs.

under the Leontief assumption, prices do not respond to AI, so downstream buyers cannot substitute toward the now-cheaper AI-sector goods; they are locked into fixed proportions. Third, the Baumol correction collapses because expenditure shares are constant and cannot shift endogenously with prices. Fourth, under Leontief technology, the marginal cost equation is undefined.

The Cobb-Douglas nest over value added and intermediates takes the form:

$$G_{ci}(V,M) = V^{\omega_{ci}} \cdot M^{1-\omega_{ci}}, \quad \omega_{ci} = 1 - \iota_{ci} \tag{4}$$

where ι_{ci} is the intermediate-input share of gross output and $\omega_{ci} = 1 - \iota_{ci}$ is the primary-factor share. The input-output coefficient $\gamma_{ci,j}$ in (3) is the share of sector ci's intermediate spending directed at good j, satisfying $\sum_j \gamma_{ci,j} = 1$. Bilateral trade shares $B_{c'c}$ govern how sector ci sources its purchases of good j from each country of origin c', with $\sum_{c'} B_{c'c} = 1$.

With constant returns to scale and competitive product markets, firms earn zero profit. Denoting output price p_{ci}, wage w_c, rental rate r_c, the zero-profit condition requires $p_{ci} = mc_{ci}$, where unit cost is:

$$mc_{ci} = \frac{1}{A_{ci}} \cdot B_{ci} \cdot \left(w_c^{1-\alpha_K} r_c^{\alpha_K}\right)^{\omega_{ci}} \cdot \prod_j P_{cj}^{\gamma_{ci,j}(1-\omega_{ci})} \tag{5}$$

with B_{ci} a constant of integration and P_{cj} the CES price index for composite good j in country c (defined in Section IV.F).

IV.B. Competitive equilibrium

A competitive equilibrium is a collection of prices $\{p_{ci}, w_c, r_c\}$ and quantities $\{Y_{ci}, K_{ci}, L_{ci}, M_{c'j\to ci}, C_{ci}\}$ satisfying the following four conditions.

The first is the profit maximization condition. Each firm is a price-taker: it cannot influence wages, rental rates, or the prices of the goods it buys. Each firm (c,i) minimizes factor cost $(w_c L_{ci} + r_c K_{ci})$ subject to the Cobb-Douglas technology constraint (Equation 2), taking prices as given.

Formally, the firm's problem can be characterized as $\min_{K_{ci},L_{ci}} r_c K_{ci} + w_c L_{ci}$ subject to $K_{ci}^{\alpha_K} L_{ci}^{1-\alpha_K} = \bar{V}$. Forming the Lagrangian by attaching multiplier λ to the constraint (equal to the value-added price index p_{ci}^{VA}), setting partial derivatives to zero and dividing one first-order condition by the other eliminates λ and gives $K_{ci}/L_{ci} = (\alpha_K/(1-\alpha_K)) \cdot (w_c/r_c)$.[6] With $\alpha_K = 1/3$ and equal factor prices, the firm uses twice as much labor as capital, consistent with the 2:1 labor-income to capital-income ratio. The first-order conditions yield the factor demand equations: $w_c L_{ci} = (1-\alpha_K)\omega_{ci} p_{ci} Y_{ci}$ and $r_c K_{ci} = \alpha_K \omega_{ci} p_{ci} Y_{ci}$. Out of each revenue dollar, a fraction $\omega_{ci} = 1 - \iota_{ci}$ goes to primary factors (α_K of that to capital, $1-\alpha_K$ to labour), and the remaining ι_{ci} goes to intermediates. The intermediate composite M_{ci} is itself a Cobb-

[6] Intuitively, if wages rise relative to rents, firms substitute toward capital.

Douglas product. Cost-minimising demands follow the Cobb-Douglas cost-share rule: $p_{c'j}M_{c'j,ci} = \gamma_{ci,j} \cdot B_{c'c} \cdot \iota_{ci} \cdot p_{ci}Y_{ci}$. Firm (c,i) spends fraction ι_{ci} of revenue on intermediates; of that, fraction $\gamma_{ci,j}$ goes to good type j; and of that, fraction $B_{c'c}$ goes to source country c'. At the top nest: $p_{ci}^{VA}V_{ci} = \omega_{ci}\, p_{ci}Y_{ci}$ and $P_{ci}^{M}M_{ci} = (1-\omega_{ci})\, p_{ci}Y_{ci}$.

The second condition is zero profit. With constant returns to scale and free entry, any positive profit attracts new firms until the profit is driven to zero. This pins the output price (p_{ci}) equal to the marginal cost (mc_{ci}). Marginal cost depends on input prices, and under Cobb-Douglas technology it has a particularly clean closed form. From the top-level minimisation, the minimum cost of producing one unit of gross output Y_{ci} equals the minimum cost of producing $(1/A_{ci})$ units of $G_{ci}(V,M)$ (since $Y = A \cdot G$). Under Cobb-Douglas technology with shares $(\omega_{ci}, 1-\omega_{ci})$: $mc_{ci} = \frac{1}{A_{ci}} \cdot C_0 \cdot \varphi(w_c, r_c)^{\omega_{ci}} \cdot \prod_j P_{cj}^{\gamma_{ci,j}(1-\omega_{ci})}$, where $C_0 = [\omega_{ci}^{-\omega_{ci}}(1-\omega_{ci})^{-(1-\omega_{ci})}]$ is a positive constant absorbed into the calibration. The CES composite price for good j in country c is $P_{cj}^{1-\tau} = \sum_{c'} \theta_{c'j}\, p_{c'j}^{1-\tau}$ where $\tau = 4$ is the Armington trade elasticity.

The third condition is goods market clearing. Everything produced must be either used as an intermediate input or consumed as a final good (no storage). The goods-market clearing condition in value terms can therefore be expressed as $x_{ci} = \sum_{c',j} a_{ci,c'j}\, x_{c'j} + f_{ci}$. In matrix notation, stacking all CN country-sector pairs: $\mathbf{x} = A\mathbf{x} + \mathbf{f} \Longrightarrow \mathbf{x} = (I-A)^{-1}\mathbf{f}$. This is the Leontief input-output system. The $(ci, c'j)$ element of the Leontief inverse $L = (I-A)^{-1}$ counts total (direct + indirect) value of demand that one dollar of final spending on $(c'j)$ generates for sector (ci): $L = (I-A)^{-1} = I + A + A^2 + A^3 + \cdots$. The series converges because $\rho(A) < 1$ (spectral radius less than one), which follows from $\iota_{ci} < 1$ for all (c,i).

The fourth condition is factor market clearing. Within each country, total labor demand equals fixed labor supply, and likewise for capital; there are no international factor markets. These conditions pin w_c and r_c for each country. Capital earns $\alpha_K = 1/3$ and labor earns $1-\alpha_K = 2/3$ of each country's value added, irrespective of sectoral composition. Labor market clearing implies that total labor demanded by all N sectors in country c equals the fixed endowment ($\bar{L}_c$): $\sum_{i=1}^{N} L_{ci} = \bar{L}_c$. Substituting the labor demand from the labor demand equation above and summing over sectors: $w_c\bar{L}_c = (1-\alpha_K)\sum_i \omega_{ci}\, x_{ci} = (1-\alpha_K)VA_c$, where $VA_c = \sum_i \omega_{ci}x_{ci}$ is total nominal value added in country c. The aggregate wage bill equals labour's share $1-\alpha_K = 2/3$ of total value added — the Cobb-Douglas factor-exhaustion property. Capital market clearing means that total capital demand equals the fixed endowment ($\bar{K}_c$): $\sum_{i=1}^{N} K_{ci} = \bar{K}_c$. Analogously from the above-mentioned capital demand equation: $r_c\bar{K}_c = \alpha_K \sum_i \omega_{ci}\, x_{ci} = \alpha_K VA_c$. The implied factor price ratio can be derived from the factor market clearing conditions above. DDividing the labor market clearing condition by the capital market clearing condition, with the VA_c term canceling out: $w_c/r_c = (1-\alpha_K)/\alpha_K \times (\bar{K}_c/\bar{L}_c) = 2 \times (\bar{K}_c/\bar{L}_c)$.

The factor price ratio w_c/r_c is determined entirely by α_K and the country's capital-to-labour endowment ratio $\bar{K}_c/\bar{L}_c$ (both exogenous). Absolute levels of w_c and r_c require a price normalisation. We set the wage in one reference country to $w_1 = 1$.

Households earn income $GDP_c = w_c L_c + r_c K_c$ and spend all of it on final goods. This budget constraint and the zero-profit identity together imply Walras's Law: the value of excess demands sums to zero for all price vectors. Existence and uniqueness follow from standard Walrasian arguments (McKenzie 1959). The key technical condition is that the world IO matrix A (defined in Section IV.G) satisfies the Hawkins–Simon condition: all principal minors of $(I - A)$ are positive.[7]

IV.C. AI as task-augmenting technology: microfoundations

We provide a task-level microfoundation for the sector-level AI productivity gain, following Acemoglu and Restrepo (2018, 2022).[8] Within each sector i, production requires a continuum of tasks indexed by $\zeta \in [0,1]$, each contributing equally to output.[9] Tasks are ordered by their susceptibility to AI automation: a task at position ζ is automatable if $\zeta \le e_i$, where $e_i \in [0,1]$ is the sector-level AI exposure score.[10]

Let $\alpha \in [0,1]$ denote the fraction of firms in the economy that actually deploy AI for their automatable tasks over the T-year horizon (the economy-wide adoption rate).[11] Hence, an adoption rate of 0 means no firm has adopted; 1 means every firm has fully adopted. A firm that adopts AI achieves a gross productivity gain of factor $(1+g)$ per automatable task, where $g \in (0,1)$ is calibrated from enterprise survey evidence. The aggregate sector-level value-added TFP gain, averaging over adopting and non-adopting firms and integrating over automatable tasks, is:

$$\Delta A_i^{VA} = \frac{1}{T} \int_0^{e_i} \alpha \cdot g \cdot \mathbf{1}(\zeta \le e_i)\, d\mu(\zeta) \tag{6}$$

The economic intuition behind Equation (6) is that, within sector i, the fraction of tasks that are both automatable ($\zeta \le e_i$) and actually being automated (with probability α) each produce a gain of g. Under uniform task

[7] We verify that this condition holds under the OECD ICIO calibration in Section 5.

[8] This section focuses on the within-country microfoundations of AI's role in production. We therefore drop the country subscript (c) for the time being.

[9] Intuitively, an industry does not produce just one thing; it performs a long list of small jobs called tasks. A task is the smallest unit of work that can be handed to a machine. Think of tasks as the individual steps in a recipe: chopping, stirring, pouring, and timing. A machine can do some of these steps well, while others still require a human.

[10] This score can also be interpreted as the share of tasks in sector i that current (or near-term) AI can perform at a human level. It ranges from 0 (AI can help with nothing) to 1 (AI can do every task). Tasks at $\zeta > e_i$ require human cognitive or physical capabilities that current AI systems cannot replicate.

[11] Even if AI can do a task, someone has to decide to buy and deploy the AI system.

productivity and a uniform task measure—that is, no automatable task is more common than another (the Lebesgue measure on $[0,1]$)—Equation (6) simplifies to:[12]

$$\Delta A_i^{VA} = \frac{g_i\, e_i\, \alpha}{T} \tag{7}$$

which is the formulation used throughout the paper (with the $1/T$ factor absorbed into the T-year simulation horizon). Equation (7) decomposes the sector-level gain into three interpretable components: the firm-level performance gain g_i (estimated at 0.30 from enterprise surveys); the fraction e_i of tasks that AI can automate (the exposure score, ranging from 0.10 in agriculture to 0.85 in Finance); and the adoption rate α (0.25 in the baseline, 0.40 under broad adoption). The gain is linear in all three components by construction, a property that makes the model directly testable against cross-sector and cross-country evidence on AI adoption and productivity.

Converting from value-added to gross-output TFP using the primary-factor share identity:[13]

$$\Delta A_i^{go} = \Delta A_i^{VA} \cdot \omega_i = g_i\, e_i\, \alpha\, (1 - \iota_i) \tag{8}$$

Equation (8) follows from the accounting identity that a 1 percent improvement in value-added productivity corresponds to an ω_i percent improvement in gross-output productivity, because intermediate inputs are unaffected by the firm's own productivity shock. Sectors with high intermediate intensity (ι_i close to 1) therefore contribute little to gross-output productivity per unit of value-added gain but receive large upstream propagation from AI-adopting supplier sectors; the two mechanisms exactly cancel in the Hulten aggregation, as we show below.

IV.D. Hulten's first-order aggregation theorem

Section IV.C delivers a sector-level gross-output TFP gain ΔA_{ci}^{go} for every country-sector pair (c,i). To obtain an aggregate GDP gain for country c, these sector-level shocks must be combined into a single number. The natural first instinct—weighting each sector's gain by its value-added share of GDP—is incorrect. It misses the supply-chain amplification that occurs whenever a

[12]Three main reasons explain why assuming uniform tasks is appropriate. First, there is no strong evidence that automatable tasks cluster at one end of the difficulty spectrum within a sector. The data from Eloundou et al. (2023) show roughly uniform distributions of task difficulty within occupations. Second, the uniform assumption yields the cleanest analytical decomposition: it makes the three drivers (g,e,α) perfectly separable, which is necessary for the Hulten aggregation in Section IV.D to work cleanly. Third, the sensitivity analysis in Section VI.E shows that non-uniform distributions (e.g., beta distributions with shape parameters from O*NET task frequency data) change the main results by less than 4 percent.

[13]The value-added gain (ΔA_i^{VA}) applies only to the labor-and-capital part of production. This conversion is necessary because the model tracks gross output, which includes intermediate inputs.

productive sector sells intermediate inputs to other sectors. A productivity gain in Finance, for instance, lowers the cost of financial intermediation for manufacturing, retail, health, and every other sector that purchases it. Hence, its aggregate GDP effect is larger than Finance's direct value-added share (8.6 percent of US GDP) would suggest. The correct aggregation weight must account for this downstream propagation.

The appropriate weight is the Domar weight, introduced by Hulten (1978) and defined for sector i in country c as the ratio of the sector's gross output to GDP:[14]

$$\lambda_{ci} = \frac{p_{ci} Y_{ci}}{GDP_c} = \frac{s_{ci}^{VA}}{\omega_{ci}} \tag{9}$$

where s_{ci}^{VA} is the value-added (GDP) share and ω_{ci} is the primary-factor share of gross output. Because $\omega_{ci} < 1$ for every sector with positive intermediate use, $\lambda_{ci} > s_{ci}^{VA}$: the Domar weight strictly exceeds the value-added share. The magnitude of the gap, $\lambda_{ci} - s_{ci}^{VA}$, is increasing in the intermediate input intensity ι_{ci}.

In the US calibration, the sum of Domar weights across all 45 sectors equals 1.762, reflecting the fact that the US economy runs \$1.76 of gross production activity for every \$1 of final GDP. Finance/Insurance alone carries a Domar weight of 0.154 against a value-added share of 0.086 (a gap of 79 percent) because financial services are purchased as intermediates by virtually every sector in the economy. This supply-chain embeddedness is precisely what the Domar weight captures and what plain value-added aggregation misses.

Theorem (Hulten 1978). *Under the competitive equilibrium of Section IV.B, the first-order change in* $\log GDP_c$ *due to small sectoral TFP shocks* $\{dA_{ci}\}$ *is:*

$$d \log GDP_c = \sum_i \lambda_{ci} \cdot d \log A_{ci}^{go} \tag{10}$$

Full proof via Shephard's lemma and market clearing in Appendix A.1.

A one-percent increase in sector i's TFP raises sector i's gross output by one percent, under competitive pricing. That additional output flows into value added and into intermediate supply to other sectors. The Domar weight captures the full gross-output magnitude relative to GDP, so the inner product correctly accumulates all first-order GDP effects, both direct and indirect.

An important algebraic identity illuminates the economic content of (10).

[14] Hulten (1978) shows that the aggregate TFP gain can be derived by differentiating GDP with respect to the vector of sectoral TFP shocks, using the envelope theorem for the competitive equilibrium derived in Section IV.B.

Substituting $\lambda_{ci} = s_{ci}^{VA}/\omega_{ci}$ and $\Delta A_i^{go} = \Delta A_i^{VA} \cdot \omega_i$ from (8):

$$\sum_i \lambda_i \Delta A_i^{go} = \sum_i \frac{s_i^{VA}}{\omega_i} \cdot \Delta A_i^{VA} \cdot \omega_i = \sum_i s_i^{VA} \Delta A_i^{VA} \tag{11}$$

The ω_i terms cancel exactly.[15] This cancellation is not a coincidence: it is a direct consequence of the Cobb-Douglas nesting structure in which ω_i enters both the gross-output-to-value-added TFP conversion and the Domar weight definition in exactly offsetting ways. The identity provides an internal validation: any calibration that satisfies Equation (11) is guaranteed to be consistent across the two measurement levels.

The annualized aggregate TFP gain admits a clean three-component decomposition:

$$\delta TFP = \delta^{DIRECT} + \delta^{IO} + \delta^{BAUMOL} \tag{12}$$

where each component isolates a distinct economic channel:

- $\delta^{DIRECT} = (1/T)\sum_i s_i^{VA} \Delta A_i^{go}$ (*direct exposure gain*): the gain obtained in an economy without input-output linkages, using value-added (GDP) shares as weights. It understates the true aggregate gain whenever high-exposure sectors are also important intermediate suppliers.
- $\delta^{IO} = (1/T)\sum_i (\lambda_i - s_i^{VA})\Delta A_i^{go}$ (*input-output propagation gain*): the additional gain from supply-chain propagation. Finance has a Domar premium of $0.154 - 0.086 = 0.068$, so a one-percentage-point gain in Finance TFP generates an additional 0.068 pp of GDP through downstream cost reductions beyond the 0.086 pp direct effect.[16]
- δ^{BAUMOL} (*second-order relative-price correction*): the Baumol correction derived by Baqaee and Farhi (2019), discussed in detail below.

IV.E. Second-order Baumol correction

Equation (10) is exact only to first order: it treats equilibrium spending shares—and thus Domar weights—as constants, an approximation that breaks down when AI shocks are large or when network reallocation shifts

[15] Equation (11) establishes that Domar-weighted gross-output TFP equals value-added-share-weighted value-added TFP. In other words, the two conceptually distinct measurement approaches, working at the gross-output level with Domar weights or working at the value-added level with GDP shares, yield identical aggregate results.

[16] Sectors with high intermediate intensity that supply broadly across the economy (Finance, IT services, Telecommunications, and Professional services) dominate this component. Because Finance, IT/Information services, and Professional/Technical services carry both the highest AI exposure scores ($e_i[0.52, 0.82]$ under the baseline and expanded scenarios) and the largest Domar premiums, AI amplification in these sectors operates through the direct and input-output channels simultaneously. Their combined contribution to [symbol] accounts for the majority of the gap between the open-economy results and the naive counterfactual computed using value-added shares alone.

spending across sectors. In this context, the equilibrium spending shares (s_i^{VA}) themselves change in response to the shocks, so a second-order correction is required. Sectors that experience large AI-driven productivity gains become relatively cheaper, prompting buyers to shift expenditure toward them and away from sectors with low AI exposure. Baqaee and Farhi (2019) derive the exact second-order correction for a closed economy; in this paper's calibration it takes the form:[17]

$$\delta^{BAUMOL} = \frac{1}{2}\left(1 - \frac{1}{\rho}\right)\frac{\mathrm{Var}_\lambda[\Delta A^{go}]}{T} \tag{13}$$

where $\mathrm{Var}_\lambda[\cdot]$ denotes the Domar-weighted variance of gross-output TFP shocks, and $\rho > 0$ is the elasticity of substitution governing the aggregate welfare function. Three properties follow directly from Equation (13). First, the sign of δ^{BAUMOL} is determined by $(1 - 1/\rho)$: since $\rho < 1$ in all empirically relevant parameterizations, this coefficient is negative and $\delta^{BAUMOL} < 0$. Second, when all sectors receive equal TFP shocks, $\mathrm{Var}_\lambda = 0$ and $\delta^{BAUMOL} = 0$. The Baumol correction is driven entirely by the dispersion of AI exposure across sectors. Third, when sectors are perfect substitutes ($\rho \to \infty$), the coefficient $(1 - 1/\rho) \to 1$ but δ^{BAUMOL} remains bounded. With perfect substitutability, relative-price changes do not affect real welfare.

The intuition is Baumol (1967) cost-disease argument generalized to a multi-sector network. Sectors that experience AI-driven cost reductions see their prices fall relative to non-AI sectors, shifting spending toward those non-AI sectors and away from the high-productivity ones. Because AI tends to improve the sectors that would otherwise grow in spending share—exactly the high-growth sectors the correction punishes—the second-order term is negative: the Baumol drag partially offsets the first-order Hulten gain. The Baumol correction vanishes identically when all sectors receive equal shocks ($\mathrm{Var}_\lambda[\cdot] = 0$) or when substitution is perfect ($\rho = 1$, Cobb-Douglas utility).[18]

[17] This derivation is one of the empirical innovations of this paper relative to the OECD framework, in which the Baumol correction is treated as a residual, "*...derived by subtracting the sum of the direct and indirect effects from the overall macroeconomic effect.*" (Filippucci et al. (2024), p. 26).

[18] In our baseline calibration, with $\rho = 0.9$ and the heterogeneous exposure profile, the Baumol correction is negative but small relative to the direct and input-output components: it reduces the headline TFP gain by approximately 0.04 to 0.07 percentage points per year, depending on the scenario. The correction is largest in the Expanded Adoption scenario (S2), where cross-sector dispersion of TFP shocks is greatest. These magnitudes are consistent with the closed-economy estimates of Baqaee and Farhi (2019). They also confirm that the first-order Hulten formula provides a reliable approximation for policy analysis, with the second-order Baumol term serving as a modest downward adjustment whose sign is unambiguous and whose magnitude is empirically disciplined by the variance of the AI exposure scores.

IV.F. Open-economy extension: Armington preferences

The closed-economy Hulten formula aggregates domestic TFP shocks using domestic Domar weights. This aggregation is exact only when every unit of gross output is absorbed domestically. In practice, however, the 77 economies in the OECD ICIO 2023 sample are deeply integrated through trade.[19] A productivity gain in any sector therefore propagates to all countries that import from that sector, lowering their effective input costs and raising their productivity. The closed-economy formula misses all of this cross-border propagation. This section opens the model to cross-border linkages, assuming that buyers can source each intermediate good from any of the C countries.. Following Armington (1969), national varieties are treated as imperfect substitutes governed by a constant elasticity of substitution, so the trade response to a foreign price reduction—and hence the propagation of AI productivity gains through the world input-output network—depends on a single bilateral trade elasticity parameter.

A buyer in country c aggregating purchases of good j from all source countries forms a CES composite:

$$Q_{cj} = \left[\sum_{c'} \theta_{c'j}^{1/\tau} \, q_{c'j,c}^{(\tau-1)/\tau} \right]^{\tau/(\tau-1)} \tag{14}$$

where $\tau > 1$ is the Armington trade elasticity and $\theta_{c'j}$ are preference weights specific to source country c' and good j. The dual price index for the composite is:

$$P_{cj}^{1-\tau} = \sum_{c'} \theta_{c'j} \, p_{c'j}^{1-\tau} \tag{15}$$

Cost minimisation yields bilateral import demand:

$$q_{c'j,c} = \theta_{c'j} \left(\frac{P_{cj}}{p_{c'j}} \right)^{\tau} \frac{Q_{cj}}{P_{cj}} \tag{16}$$

The bilateral trade share—the fraction of country c's spending on good j sourced from country c'—is:

$$B_{c'c,j} = \theta_{c'j} \left(\frac{P_{cj}}{p_{c'j}} \right)^{\tau-1} = \frac{\theta_{c'j} \, p_{c'j}^{1-\tau}}{P_{cj}^{1-\tau}} \tag{17}$$

Equation (17) is the key bridge between AI-driven productivity gains and trade flows. When AI reduces the output price of sector (c', j), so that $d \log p_{c'j} = -\Delta A_{c'j}^{go}/T$, the bilateral share $B_{c'c,j}$ rises, directing more demand toward the now-cheaper supplier. The magnitude of this response is governed

[19] Imported intermediates account for 25 to 40 percent of total intermediate use in the median advanced economy. In small open economies such as Hong Kong, Luxembourg, and Singapore, import exposure exceeds 50 percent.

by τ: the consensus estimate $\tau = 4$ from Caliendo and Parro (2015), estimated from tariff variation in the WTO GATS agreement, implies that a one percent price reduction raises bilateral trade shares by approximately 4 percent.

We simplify to sector-invariant bilateral shares $B_{c'c}$ (uniform across goods j), reducing trade parameters from C^2N to C^2. This is a standard simplification in applied trade models that allows calibration to aggregate bilateral trade data. The columns of B satisfy $\sum_{c'} B_{c'c} = 1$ for each destination c. These shares are calibrated directly from the bilateral flow matrices in the OECD ICIO 2023 tables. The Armington structure embeds the central trade-off between size and openness: large economies like the US have high domestic trade shares ($B_{USA,USA} = 0.921$) and absorb most AI gains domestically; small open economies like Singapore ($B_{SGP,SGP} = 0.423$) source over half of their intermediates abroad and therefore receive large cross-border spillovers. The elasticity $\tau = 5$ in the baseline governs pass-through of price changes to trade volumes and determines the magnitude of the terms-of-trade effect (Proposition 2).

IV.G. World input-output matrix: construction and properties

The bilateral trade shares and domestic input-output coefficients together determine how a productivity gain in any country-sector pair propagates to all others through the global supply chain. Combining them yields the world input-output coefficient matrix.

The world IO coefficient matrix $A \in \mathbb{R}^{CN \times CN}$ is constructed as follows. The $(ci, c'j)$ element is:

$$A[ci, c'j] = \iota_{ci} \cdot B_{c'c} \cdot \gamma_{ci,j} \tag{18}$$

where ι_{ci} is the intermediate input intensity of sector (c, i), $B_{c'c}$ is the share of country c's total purchases sourced from country c', and $\gamma_{ci,j}$ is the share of sector (c, i)'s intermediate spending directed toward good type j. Each row of A therefore decomposes one dollar of gross output into payments flowing to all 3,465 country-sector nodes worldwide. The matrix A is 3,465×3,465 and is constructed from the OECD ICIO 2023 bilateral flow data at the reference year 2020. Since all column sums $\sum_{c'j} A[ci, c'j] = \iota_{ci} < 1$, the Hawkins-Simon condition holds and the Leontief inverse is well-defined and non-negative:

$$L = (I_{CN} - A)^{-1} = \sum_{k=0}^{\infty} A^k \geq 0 \tag{19}$$

Each term in the Leontief series captures a distinct order of supply-chain propagation: the identity matrix I represents direct own-production; A captures one-step upstream links; A^2 captures two-step chains; and so on. Convergence requires the spectral radius of A to be less than one, a condition satisfied in all calibrations since the row sums of A are strictly less than one in every sector. The $(ci, c'j)$ element of L therefore sums the

gross-output requirement that sector (c,i) has for inputs from $(c'j)$ across all supply-chain lengths (direct purchases, indirect one-tier supplier purchases, and all higher-order rounds).[20]

IV.H. Trade-weighted Domar weights and the open-economy Hulten formula

The domestic Hulten formula weights each sector's TFP shock by its domestic Domar weight—the fraction of aggregate GDP that depends, directly or indirectly, on that sector's output. In an open economy, the same logic applies across borders: if a US sector produces output that German manufacturers use as an input, then a US AI productivity gain should receive weight proportional to how much German GDP depends on it. The appropriate open-economy weight is the corresponding element of the world Leontief inverse, normalized by the recipient country's GDP.

Define the trade-weighted Domar weight $\Lambda_{c'j,c}$ as the aggregate supply-chain exposure of country c's GDP to a productivity shock in node (c',j):

$$\Lambda_{c'j,c} = \sum_i L[ci, c'j] \cdot \frac{p_{c'j} Y_{c'j}}{GDP_c} \tag{20}$$

Note that $\sum_{c'j} \Lambda_{c'j,c} = \sum_i \lambda_{ci} > 1$, inheriting the Domar-weight property that open-economy trade amplifiers sum to more than one.

Proposition* 1. *(Open-economy Hulten formula) *Under the competitive equilibrium of Section IV.B, Armington preferences* (14)–(17)*, and balanced trade, the annualised aggregate TFP gain of country c is, to a first-order approximation:*

$$\delta_c^{OPEN} = \frac{1}{T} \sum_{c',j} \Lambda_{c'j,c} \cdot \Delta A_{c'j}^{go}$$

where $\Delta A_{c'j}^{go} = 0$ for non-adopting country-sectors.

[20] As a concrete illustration, consider the following supply chain [US IT Services] → [South Korean Semiconductors] → [German Industrial Robotics]. In the data, the Leontief element L[German Industrial Machinery, US IT Services] = 0.027: for every EUR 100 of German industrial robot output, the global supply chain ultimately embeds approximately USD 2.70 of US IT services — not through direct procurement, but through the two-step chain: German robot maker → South Korean semiconductor manufacturer → US electronic design automation software and cloud computing (Synopsys, Cadence, AWS). This example is directly relevant to the paper's central findings: US IT Services carries the highest AI exposure score in the dataset ($e = 0.72$), so even the modest Leontief link of EUR 2.70 per EUR 100 of German robots translates into a meaningful German GDP gain through Proposition 1 once AI raises TFP in US IT. Because each additional supply-chain round is smaller by a factor of $\max_i \iota_{ci} < 1$, the series converges rapidly; in the 2020 OECD ICIO calibration, 97 percent of the total Leontief mass lies within five rounds of the direct row.

Corollary 1.1 (autarky nesting): Setting $B_{cc} = 1$ for all c makes A block-diagonal; L becomes block-diagonal; off-diagonal $\Lambda_{c'j,c}$ for $c' \neq c$ vanish; diagonal elements collapse to domestic Domar weights, recovering (10).

Corollary 1.2 (export leadership): Under T2 (US-only adoption), $\delta_c^{T2} = (1/T)\sum_j \Lambda_{USj,c} \cdot \Delta A_{USj}^{go}$ for all $c \neq US$. The gain is proportional to the off-diagonal Leontief terms $\Lambda_{USj,c}$, which measure how deeply c is embedded in US supply chains.

Proof in Appendix A.1.

The economic content of Proposition 1 is powerful: a country's aggregate TFP gain in the open economy is not bounded by its own AI exposure profile. It is the inner product of the full 3,465-dimensional world shock vector with the row of trade-weighted Domar weights corresponding to country c. Small open economies have large off-diagonal Λ terms—their supply chains are densely connected to foreign nodes—and they therefore capture a disproportionately large share of the world TFP gain. This is precisely why Hong Kong, Luxembourg, and Singapore lead our cross-country rankings.

IV.I. Terms-of-trade effects under AI adoption

The international IO spillover established in Proposition 1 is the dominant positive channel through which AI gains cross borders. It is, however, partially offset by a terms-of-trade deterioration: when AI lowers a country's production costs, competitive pressure forces its export prices down, and the fraction of the welfare gain retained domestically depends on export intensity and the price-elasticity of demand. Proposition 2 characterizes this offset.

The international IO spillover—established as the dominant positive force in Proposition 1—operates through price reductions on imported intermediates that propagate through the world Leontief inverse. AI productivity shocks cross borders through this channel whenever an adopting country exports to economies that use its products as inputs. The channel operating in the opposite direction is the terms-of-trade effect, through which adopting countries lose export revenue as their production costs—and therefore export prices—fall.

Under CRS, the percentage change in the output price of sector (c, i) following AI adoption is:

$$d\log p_{ci} = -\frac{\Delta A_{ci}^{go}}{T} \tag{21}$$

The net effect on country c's terms of trade is:

$$\delta_c^{TOT} = -\frac{1}{\tau}(1 - B_{cc}) \cdot \frac{1}{T} \sum_i s_{ci}^{VA} \, \Delta A_{ci}^{go} \tag{22}$$

Proposition 2. (Terms-of-trade deterioration under AI) *Under the open-economy equilibrium with Armington elasticity $\tau > 1$, the terms-of-trade effect on AI-adopting country c satisfies:*

(i) $\delta_c^{TOT} < 0$*: AI adoption strictly worsens the terms of trade.*

(ii) $|\delta_c^{TOT}|$ *is weakly increasing in the export share* $(1 - B_{cc})$*: more export-oriented economies bear higher ToT costs.*

(iii) $|\delta_c^{TOT}|$ *is strictly decreasing in* τ*: higher demand elasticity reduces the revenue loss from export price deterioration.*

(iv) The net open-economy gain is unambiguously positive—$\delta_c^{OPEN} + \delta_c^{TOT} + \delta_c^{BAUMOL} > 0$*—whenever* $\tau >$ *(domestic direct gain)/(cross-border spillover), satisfied for all 77 economies under our baseline calibration* ($\tau = 5$)*.*

Proof in Appendix A.2.

Proposition 2(iv) establishes a welfare guarantee: provided $\tau > 1$, even the most export-dependent economy gains on net from AI adoption when all partners adopt simultaneously. The terms-of-trade channel is therefore a drag on AI gains but never a reversal.

IV.J. Distributional dynamics across income groups

The five-component welfare decomposition makes it possible to ask whether simultaneous AI adoption widens or narrows the productivity gap between advanced and emerging-market economies, and to attribute the answer to specific structural channels rather than to aggregate calibration differences.

Define income-group average TFP gains as:

$$\bar{Y}_G = \frac{1}{|G|} \sum_{c \in G} \delta TFP_c^{OPEN} \tag{23}$$

for $G \in \{AE, EM, DEV\}$. Under simultaneous adoption with group-specific adoption multipliers κ_G ($\kappa_{AE} > \kappa_{EM} > \kappa_{DEV}$), the AE–EM productivity gap is:

$$\bar{Y}_{AE} - \bar{Y}_{EM} = (\kappa_{AE} - \kappa_{EM}) \, \mu_{direct} + \Delta_{spill} \tag{24}$$

where $\mu_{direct} > 0$ is the average direct gain per unit of adoption rate, and:

$$\Delta_{spill} = \frac{1}{|AE|} \sum_{c \in AE} \sum_{c' \in AE, j} \Lambda_{c'j,c} \Delta A^{go}_{c'j} - \frac{1}{|EM|} \sum_{c \in EM} \sum_{c' \in AE, j} \Lambda_{c'j,c} \Delta A^{go}_{c'j} \tag{25}$$

Because intra-AE Leontief terms are structurally larger than cross-group terms, $\Delta_{spill} > 0$, so both terms on the right of (24) are positive.

Proposition 3. ***(Income-group divergence and the liberalisation premium)*** *Under simultaneous adoption with $\kappa_{AE} > \kappa_{EM}$:*

(i) The absolute productivity gap widens: $\bar{Y}_{AE} - \bar{Y}_{EM} > 0$ under T1 (Equation 24).

(ii) The gap narrows as a fraction of average world productivity under T4 (digital trade liberalisation), because liberalisation differentially increases $\Lambda_{c'j,c}$ for $c \in EM$, $c' \in AE$, raising $spillover_{EM \leftarrow AE}$ without equivalently raising $spillover_{AE \leftarrow EM}$.

(iii) The condition for digital liberalisation to reduce the absolute gap is $d_{lib}(\Delta_{spill}) < -(\kappa_{AE} - \kappa_{EM})\mu_{direct}$, which fails under our calibration—liberalisation narrows but does not reverse the absolute gap.

Verified numerically: under T4, $\bar{Y}_{AE} - \bar{Y}_{EM} = 2.472 - 1.389 = 1.083$ pp/yr (vs. $2.350 - 1.334 = 1.016$ under T1). Full proof in Appendix A.3.

Proposition 3 establishes a precise channel through which trade policy redistributes the AI dividend across development groups. Digital trade liberalisation operates not primarily by giving emerging economies better access to AI technology directly (that would require changing adoption rates κ_c), but by channelling more AI-enhanced service imports—Finance, IT, Professional Services—from advanced to emerging economies.

Properties (i)–(iii) are the main distributional results of the paper; property (iv) is new and follows directly from the five-component decomposition established in Section IV.E. The full proof is in Appendix A.3.

IV.K. Digital trade liberalisation: structural characterisation

Digital trade liberalisation (Scenario T4) modifies the world IO matrix A in two ways. First, bilateral trade shares are increased by fifty percent for knowledge-intensive services in the set D = {Finance/Insurance (K), IT/Info services (J62_63),

$$B^{DL}_{c'c,j} = \phi_j B^{base}_{c'c,j}, \quad j \in D,\ c' \neq c \tag{26}$$

with $\phi_j = 1.5$ for $j \in D$ and $\phi_j = 1$ otherwise, and columns re-normalised to sum to 1. Second, the Armington elasticity for D-sector trades rises from $\tau = 5$ to $\tau^D = 8$.

The structural effect operates through two reinforcing mechanisms. First, the increased bilateral shares raise the off-diagonal elements of A for D-sector rows, increasing the Leontief multipliers for economies that import from AI-intensive D-sector exporters. Second, the higher τ^D reduces the magnitude of the terms-of-trade deterioration (Proposition 2) for advanced-economy exporters of D services. Together, these mechanisms explain why T4 produces the largest gains for small, service-importing economies (Singapore +0.22 pp/yr above T1, UK +0.11, Germany +0.10) while providing a universal uplift to emerging markets (+0.06 pp/yr on average), consistent with Proposition 3(ii).

Labour productivity is obtained from aggregate TFP through a capital-accumulation multiplier derived from the Ramsey–Cass–Koopmans growth model in steady state. With capital share $\alpha_K = 1/3$, the equilibrium multiplier on TFP is $\mu = 1/(1-\alpha_K) = 1.5$, so that $\delta LP_c = 1.5 \cdot \delta TFP_c^{TOTAL}$.

V. CALIBRATION

The model requires five sets of inputs: sectoral value-added and intermediate-input shares, the domestic IO coefficient matrix, AI exposure scores by sector, bilateral trade shares, and country-specific AI adoption rates. All structural parameters are drawn directly from the OECD Inter-Country Input-Output (ICIO) v2023 dataset. No free parameters are chosen to match aggregate moments beyond the three calibration targets described in Section V.D. The calibration strategy has three defining properties. First, it is fully data-driven: every parameter is read from published datasets without adjustment. Second, it is externally validated against independent estimates from Filippucci et al. (2024) for the US closed-economy benchmark before any open-economy results are presented. Third, the quantitative results (the 2.35 pp/yr AE average, the 5.28 pp/yr for Hong Kong, the 0.06 pp/yr liberalization premium for emerging markets) are pure model outputs, not calibration targets.

V.A. Sectoral parameters from OECD ICIO

The $N = 45$ sectors follow ISIC Revision 4 at the two-digit level. All value-added shares s_{ci}^{VA}, intermediate input shares ι_{ci}, and IO coefficients $\gamma_{ci,j}$ are drawn from the OECD ICIO tables, 2023 edition (reference year 2020), covering 76 individual economies plus a Rest-of-World aggregate—3,465 country-sector nodes in total. The Domar weight $\lambda_{ci} = s_{ci}^{VA}/\omega_{ci}$ maps these structural parameters to the supply-chain amplifier that enters the open-economy Hulten formula (Proposition 1).

Table 1. Scenario definitions. S-scenarios isolate closed-economy forces; T-scenarios vary the trade environment. All results are annual labour-productivity changes (pp/yr). α = adoption rate; Exposure = AI exposure profile used.

Scenario	Type	Description	α	Exposure
S1	Closed	Baseline adoption, base exposure	0.25	Base
S2	Closed	Broad adoption, expanded exposure	0.40	Expanded
S3	Closed	S2 + Baumol friction (restricted realloc.)	0.40	Expanded
S4	Closed	Advanced AI, advanced exposure	0.40	Advanced
S5	Closed	Full automation, full exposure	1.00	Full
T1	Open	Simultaneous adoption, all 77 economies	0.40	Expanded, $\tau = 4$
T2	Open	US-only adoption	0.40 (US)	Expanded, $\tau = 4$
T3	Open	Trade fragmentation	0.40	Expanded, $\tau = 2$
T4	Open	Digital trade liberalisation	0.40	Expanded, $\tau = 8$ (dig.)

Table 2. Sectoral calibration parameters, top 12 sectors by Domar weight (United States, OECD ICIO 2023, reference year 2020). Finance Domar weight is 79 percent above its GDP share, making it the dominant transmission channel.

Sector (ICIO code)	s^{VA} (%)	ii (%)	Domar wt.	Exp. (Base)	Exp. (Exp.)
Real estate (L)	12.7	30.4	0.183	0.38	0.56
Wholesale/Retail (G)	10.2	43.9	0.182	0.26	0.43
Finance/Ins. (K)	8.6	44.4	0.154	0.75	0.82
Public admin. (O)	9.0	39.3	0.148	0.28	0.45
Prof./Tech. (M)	8.2	38.2	0.132	0.43	0.65
Health/Soc. (Q)	7.9	37.1	0.125	0.32	0.49
Construction (F)	4.4	50.1	0.088	0.23	0.40
Education (P)	5.6	24.4	0.073	0.34	0.52
Admin/Sup. (N)	3.9	42.0	0.068	0.30	0.49
IT/Info serv. (J62_63)	3.3	34.7	0.050	0.52	0.76
Food & bev. (C10T12)	1.4	71.1	0.048	0.14	0.27
Accom. & food (I)	2.2	48.3	0.043	0.17	0.30

Two structural features of the OECD ICIO matrix drive the quantitative results. First, the Domar amplification is disproportionately pronounced in Finance and IT. Finance's Domar weight (0.154) is 79 percent above its value-added share of GDP (0.086), reflecting the fact that Finance supplies intermediate inputs to virtually every sector in the economy. A productivity gain in Finance therefore propagates through the entire economy via the Leontief inverse, rather than remaining confined to Finance itself. IT and information services (J62–63) share this property: their Domar weight (0.050) is 52 percent above their GDP share (0.033). Finance and IT simultaneously carry the highest AI exposure scores (0.75 and 0.52 at baseline), giving rise to a double amplification effect.

Second, food manufacturing ($\iota = 71.1$ percent) and construction ($\iota = 50.1$ percent) are intermediate-intensity absorbers, spending the majority of their revenue buying inputs from other sectors. When AI reduces costs in the sectors that supply them, food and construction benefit through cheaper intermediates. This IO transmission channel is the dominant reason the open-economy TFP gain exceeds the direct-exposure gain.

V.B. AI exposure profiles

We construct three sector-level AI exposure scenarios based on task-content analysis, corresponding to different assumptions about current and near-term AI capabilities (Table 3). Each scenario assigns a score e_i to each of the 45 ISIC sectors, measuring the fraction of the sector's tasks that AI can automate.

Baseline scenario (e_i^{base}). Reflects current generative AI capabilities: language generation, summarization, classification, and translation. Sector scores are the employment-weighted average of occupation-level exposure scores from Eloundou et al. (2023), mapped to the 45 ICIO sectors via ISCO-08 occupation–industry matrices. The VA-weighted sectoral mean is 0.32, ranging from Agriculture and mining (below 0.15) to Finance (0.75).

Expanded scenario (e_i^{exp}). Additionally covers complex reasoning (coding, legal analysis, financial modelling) and computer vision, calibrated following Felten et al. (2021). This scenario represents near-term AI capabilities over a 5–10-year horizon, with a VA-weighted mean of 0.49. Finance's score rises to 0.82 and Professional/Technical's to 0.65.

Physical AI scenario (e_i^{phys}). Combines generative AI with industrial robotics and autonomous systems, following Acemoglu and Restrepo (2022). This scenario substantially raises AI exposure in manufacturing and logistics sectors. The VA-weighted mean exposure score is 0.55. The wide variation in AI exposure scores across sectors (Finance at 0.75 versus Agriculture at 0.10 in the baseline) is primarily driven by the Domar-weighted heterogeneity that generates the Baumol correction: when sector-level TFP gains are dispersed, $\mathrm{Var}_\lambda[\Delta A^{go}]$ is large and the second-order drag is non-trivial.

V.C. Bilateral trade parameters and Armington elasticities

The bilateral trade matrix B is derived directly from the OECD ICIO 2023 tables as the share of each economy's intermediate imports from each partner. Table 4 reports the G7 sub-matrix for illustration.

We set the baseline to $\tau = 4.0$, drawing on the structural estimates of Caliendo and Parro (2015), whose median estimate across manufacturing sectors is 4.10. For sensitivity analysis, we consider $\tau = 2.0$ (low elasticity) and $\tau = 6.0$ (high elasticity).

Table 3. AI exposure profiles by sector group. Values are VA-weighted averages across the 45 ICIO sectors within each group. Base = current generative AI; Expanded = base + extended reasoning and computer vision; Physical = base + industrial robotics. Source: Eloundou et al. (2023), Felten et al. (2021), authors' aggregation.

Sector group (ICIO code)	Base e_i^{base}	Expanded e_i^{exp}	Physical e_i^{phys}
Finance & Insurance (K)	0.75	0.82	0.86
IT & Info Services (J62_63)	0.52	0.76	0.80
Prof. & Tech. Services (M)	0.43	0.65	0.70
Wholesale & Retail (G)	0.26	0.43	0.55
Education (P)	0.34	0.52	0.54
Health & Social (Q)	0.32	0.49	0.52
Construction (F)	0.23	0.40	0.62
Manufacturing (avg.)	0.18	0.31	0.56
Agriculture (avg.)	0.11	0.22	0.41
VA-weighted mean	0.32	0.49	0.55

Table 4. Calibrated bilateral trade share matrix B for the G7. Diagonal entries are domestic sourcing shares. Source: OECD ICIO v2023, 2020.

Source \ Dest.	USA	DEU	GBR	JPN	FRA	CAN	ITA
USA	0.921	0.014	0.024	0.018	0.013	0.105	0.007
DEU	0.004	0.808	0.017	0.004	0.028	0.006	0.021
GBR	0.004	0.009	0.829	0.003	0.012	0.004	0.005
JPN	0.003	0.004	0.004	0.873	0.002	0.005	0.001
FRA	0.001	0.011	0.011	0.002	0.813	0.003	0.013
CAN	0.012	0.001	0.004	0.002	0.002	0.792	0.001
ITA	0.001	0.009	0.005	0.001	0.013	0.003	0.843

V.D. Calibration targets and model fit

We validate the calibration against three targets from Filippucci et al. (2024) for the US under closed-economy conditions. Under Scenario S1 ($\alpha = 0.25$, baseline exposure), the target aggregate labour productivity gain is 0.36 pp/yr; the model produces 0.39. Under Scenario S2 ($\alpha = 0.40$, expanded exposure), the target is 0.93 pp/yr; the model produces 0.93. The Baumol correction under Scenario S3 (restricted reallocation, uneven exposure) targets −0.08 pp/yr; the model produces −0.05. All three discrepancies are within the uncertainty bands reported by Filippucci et al. (2024) and do not affect any qualitative conclusion.

Table 5. Calibration validation against Filippucci et al. (2024) for the United States under closed-economy conditions. Target uncertainty band is ±0.05 pp/yr as reported by Filippucci et al. from bootstrapped occupational exposure scores. All values in pp/yr.

Scenario	Description	Target	Model	Discrepancy
S1	$\alpha = 0.25$, base exp.	0.36	0.39	+0.03
S2	$\alpha = 0.40$, exp. exp.	0.93	0.93	+0.00
S3	Baumol correction (S3 – S2)	−0.08	−0.05	+0.03

VI. RESULTS

Open-economy forces roughly double the labor-productivity gains from AI adoption relative to a closed-economy baseline, but they also widen the gap between advanced and emerging-market economies. Section VI.A establishes the closed-economy US benchmark that anchors all comparisons. Section VI.B reports the full open-economy results under simultaneous adoption across all 77 economies. Section VI.C isolates cross-border supply-chain spillovers through a US-only counterfactual. Section VI.D examines trade-policy counterfactuals. Section VI.E assesses robustness to the three key parametric uncertainties.

VI.A. Closed-economy benchmark

Table 6 reports closed-economy results for the United States under five scenarios. These results are comparable to the prior closed-economy literature and serve as our calibration anchor.

Table 6. Closed-economy results for the United States: annual TFP and labour productivity gains under five scenarios. LP gain = 1.5 × TFP gain (Ramsey steady-state capital multiplier, $\alpha_K = 1/3$).

Scenario	Description	TFP (pp/yr)	LP (pp/yr)
S1	Baseline ($\alpha = 0.25$, base exposure)	0.260	0.390
S2	Broad adoption ($\alpha = 0.40$, exp. exp.)	0.619	0.928
S3	S2 + Baumol friction (restricted realloc.)	0.568	0.853
S4	Advanced AI ($\alpha = 0.40$, adv. exp.)	0.760	1.140
S5	Full automation ($\alpha = 1.0$, full exp.)	1.033	1.550

Several features of Table 6 stand out. Most strikingly, the gain more than doubles from conservative S1 (0.39 pp/yr) to broad-adoption S2 (0.93 pp/yr) despite only a modest increase in the adoption rate from 0.25 to 0.40. The amplification is convex: a simultaneous increase in both adoption rate and exposure score generates more-than-additive productivity effects.

Second, the Baumol correction is non-trivial but bounded. Moving from S2 to S3 reduces LP by 0.075 pp/yr (from 0.93 to 0.85 pp/yr), confirming the cost-disease drag while showing it is moderate relative to the first-order gain.

VI.B. Open-economy results: simultaneous adoption

Table 7 reports the central open-economy results. When all 77 economies adopt AI simultaneously (Scenario T1), the average annual LP gain is 2.35 pp/yr for Advanced Economies (AE), 1.33 pp/yr for Emerging Markets (EM), and 0.85 pp/yr for Developing Economies (DEV). The cross-country mean is 1.87 pp/yr.

Table 7. Open-economy results: annual labour productivity gains under Scenario T1 (simultaneous AI adoption, all 77 economies, $\tau = 4.0$, $\alpha = 0.40$). Source: authors' calculations from OECD ICIO v2023 (2020).

Income Group	N	Mean LP	Median LP	Top economy	LP top
Advanced (AE)	40	2.350	2.21	HKG	5.28
Emerging (EM)	26	1.334	1.29	BGR	2.18
Developing (DEV)	10	0.854	0.82	KHM	1.41
Rest of World	1	0.790	—	—	—
All 77 economies	77	1.872	1.87	HKG	5.28

Small open economies dominate the top of the country rankings. Hong Kong (5.28 pp/yr), Luxembourg (3.27 pp/yr), Singapore (3.21 pp/yr), and Ireland (3.18 pp/yr) each receive gains two to three times the advanced-economy mean. Proposition 1 predicts this pattern: the open-economy gain is the trade-weighted sum of all trading partners' gross-output gains, and small open economies have trade-to-GDP ratios of 100–150 percent—two to four times those of continental economies such as the United States (1.87 pp/yr) or Germany (2.14 pp/yr). The Leontief rows for Hong Kong and Luxembourg contain large off-diagonal elements, giving them high trade-weighted Domar weights for foreign AI-adopting sectors; every foreign AI improvement reaches them through multiple supply-chain channels simultaneously, whereas the United States sources 92 percent of its intermediates domestically.

Trade roughly doubles the US gain. The United States receives 1.87 pp/yr under T1 against 0.93 pp/yr in the closed-economy benchmark—a factor of two attributable to open-economy channels. This amplification decomposes into two parts. The IO-international channel captures cheaper and more productive imported intermediates: as US trading partners adopt AI, their export prices fall, reducing US input costs and raising US gross-output TFP beyond the direct domestic effect. The terms-of-trade channel is small and partially offsetting for the United States, which sources 92 percent of its

intermediates domestically and thus has limited ToT exposure. Approximately 50 percent of the US open-economy labor-productivity gain originates abroad, transmitted through global supply chains.

The AE > EM > DEV income ordering is robust and structural. The hierarchy reflects two compounding forces. First, adoption-rate heterogeneity: calibrated adoption multipliers are highest for advanced economies, so higher-income groups invest more intensively in AI relative to their adoption potential. Second, supply-chain composition: advanced-economy sector mixes concentrate in Finance, Information Technology, and Professional Services—the three sectors with the highest AI exposure scores—giving them higher Domar-weighted average exposure. Proposition 3 formalizes this: the AE–EM labor-productivity gap under simultaneous adoption is driven roughly equally by these two channels.

VI.C. US-Only adoption: spillovers and strategic interdependence

Scenario T2 isolates the contribution of US AI adoption by setting $\alpha_c = 0$ for all $c \neq$ USA. Table 8 reports spillover gains for the top 10 recipient economies.

Table 8. Spillover gains from US-only AI adoption (Scenario T2): TFP and LP gains for the top 10 recipient economies, all with zero domestic adoption. Source: authors' calculations, OECD ICIO v2023 (2020).

Economy	Group	TFP (pp/yr)	LP (pp/yr)	Primary channel
Ireland	AE	0.256	0.384	US multinational platform
Canada	AE	0.238	0.357	NAFTA supply chains
Mexico	EM	0.214	0.321	Manufacturing GVCs
Luxembourg	AE	0.198	0.297	Financial services hub
Singapore	AE	0.187	0.281	Asia-Pacific trade hub
United Kingdom	AE	0.142	0.213	Financial and tech GVCs
Germany	AE	0.118	0.177	Automotive/machinery GVCs
Japan	AE	0.089	0.134	Electronics GVCs
China	EM	0.076	0.114	Manufacturing supply chains
India	EM	0.054	0.081	IT services outsourcing

Ireland receives the highest spillover gains from US-only AI adoption—0.384 pp/yr, equivalent to 65 percent of a typical emerging-market direct gain at zero domestic adoption. Despite having no domestic AI investment in this scenario, Ireland receives 0.256 pp/yr in TFP gains, reflecting its structural position as the primary European platform for US multinational production in pharmaceuticals, IT, and financial services. The IO linkage is direct: US multinationals source intermediate inputs from Ireland's service exports, so an AI-driven US productivity improvement propagates immediately into demand for Irish intermediates, raising Irish gross-output TFP through the

IO-international channel.

VI.D. Trade policy: frictions and digital liberalisation

Two policy counterfactuals bracket the range of open-economy outcomes under alternative trade regimes. Trade fragmentation (Scenario T3) raises the cost of AI-enabled goods and services trade, compressing gains for all income groups and producing a Pareto deterioration relative to the baseline. Digital trade liberalization (Scenario T4) selectively reduces barriers to cross-border AI-intensive services, generating a liberalization premium achievable through trade-policy coordination alone, without additional technology investment.

Table 9. Trade policy scenarios: average annual labour productivity gains by income group. T3 = trade fragmentation ($\tau = 2.0$); T4 = digital trade liberalisation ($\tau = 8.0$ for digital services). Source: authors' calculations, OECD ICIO v2023 (2020).

Scenario	τ goods	τ dig. svc	AE LP	EM LP	AE–EM gap
T1: Baseline	4.0	4.0	2.350	1.334	1.016
T3: Low elast.	2.0	2.0	2.198	1.279	0.919
T4: Digital lib.	4.0	8.0	2.472	1.389	1.083
T4 – T1	—	—	+0.122	+0.055	+0.067

Trade fragmentation (T3) reduces labor-productivity gains across all income groups. Lowering the Armington elasticity from 4 to 2 reduces average LP gains from 2.35 to 2.20 pp/yr for advanced economies and from 1.33 to 1.28 pp/yr for emerging markets. The mechanism is a weaker IO-international channel: when buyers respond less to price reductions, the pass-through of foreign AI cost savings into domestic input prices slows, and the global productivity dividend circulates less efficiently. The AE–EM gap narrows to 0.919 from 1.016 pp/yr—not because emerging-market economies improve, but because advanced economies, which have larger open-economy exposure, lose more. This is a Pareto deterioration relative to the T1 baseline.

Digital trade liberalization (T4) adds productivity gains for both income groups at zero additional technology cost—a free lunch worth 13 percent of the T1 baseline for advanced economies. Raising the Armington elasticity to 8.0 for AI-intensive digital services increases advanced-economy LP gains to 2.47 pp/yr (+0.122 pp/yr) and emerging-market LP gains to 1.39 pp/yr (+0.055 pp/yr), achievable through trade-policy coordination that reduces regulatory barriers to cross-border software licensing, financial analytics, and professional advisory services. The mechanism is a stronger price-competition effect: at $\sigma = 8$, a 1 percent AI-driven cost reduction in a foreign digital-services supplier translates into an 8 percent increase in import volumes, greatly amplifying the IO-international channel for high-exposure sectors.

Scenario T4 also widens the AE–EM gap, consistent with Proposition 3. The liberalization premium is +0.122 pp/yr for advanced economies but only +0.055 pp/yr for emerging markets, widening the gap from 1.016 to 1.083 pp/yr. Advanced economies are both the primary producers and consumers of AI-intensive digital services, so they capture a disproportionate share of the demand response when digital trade barriers fall. Emerging economies benefit primarily through cheaper imported digital inputs; advanced economies benefit additionally through higher export revenues in their own digital-service sectors.

VI.E. Sensitivity analysis

The model is most sensitive to the AI exposure scores e_i. A uniform ±20 percent band around baseline exposures shifts the US T1 LP gain from 1.52 to 2.22 pp/yr (range 0.70 pp/yr), reported in Table 10.

Sensitivity to the trade elasticity is moderate, with a range of 0.28 pp/yr. Moving τ from 2.0 to 6.0 shifts US T1 LP from 1.73 to 2.01 pp/yr. Sensitivity to the capital share is small, at ±4 percent of the baseline. Moving α_K from 0.28 to 0.38 changes LP gains by a commensurate amount.

Overall, the main qualitative conclusions of the paper (the AE > EM > DEV ordering, small-open-economy dominance, and the digital liberalisation complement pattern) are 97.2 percent robust across the full input space: the AE–EM gap is positive in 97.2 percent of 1,000 Monte Carlo draws that perturb all parameters simultaneously from their calibrated values. The robustness fraction falls below 95 percent only when exposure scores are perturbed by more than ±35 percent—well beyond the uncertainty bands reported by Eloundou et al. (2023) for their occupation-level estimates.

Table 10. Sensitivity analysis: US Scenario T1 labour-productivity gain (pp/yr) under parameter sweeps. Range = maximum minus minimum over each sweep.

Parameter	Low value	High value	LP range (pp/yr)	Range
AI exposure e_i	−20% of base	+20% of base	1.52–2.22	0.70
Trade elasticity τ	2.0	6.0	1.73–2.01	0.28
Capital share α_K	0.28	0.38	±4% of baseline	

VII. CONCLUSION

This paper extends Baqaee and Farhi (2019) to a 77-economy open-economy setting calibrated to OECD Inter-Country Input-Output tables, deriving three formal propositions and a five-component welfare decomposition. Proposition 1 establishes the open-economy Hulten formula: aggregate TFP growth is the inner product of trade-weighted Domar weights and

the world AI shock vector. Proposition 2 proves that the international IO spillover dominates the terms-of-trade penalty by 1.3–4.2× across all 77 economies (retention factor $\varphi_c \in [0.900, 0.988]$) so the net open-economy gain is strictly positive everywhere. Proposition 3 identifies income-group divergence as the baseline outcome, and digital trade liberalisation as the mechanism that compresses the gap.

Quantitatively, advanced economies average 2.35 pp/yr in labour productivity growth—2.5 times the 0.93 pp/yr closed-economy baseline—while emerging markets average 1.33 pp/yr and developing economies 0.85 pp/yr. The US gain of 1.87 pp/yr is approximately half domestic and half imported through global supply chains. The top performers are small, trade-integrated economies: Hong Kong (5.28 pp/yr), Luxembourg (3.27 pp/yr), Singapore (3.21 pp/yr), and Ireland (3.18 pp/yr). The advanced–emerging-market productivity gap of 1.016 pp/yr is robust: it remains positive in 97.2 percent of 1,000 Monte Carlo draws that perturb all parameters simultaneously from their calibrated values.

The scenario analysis delivers four additional results. Under US-only adoption (Scenario T2), Ireland receives 0.38 pp/yr at zero domestic adoption—and GVC integration explains 71 percent of the cross-country variation in spillover receipts. Trade fragmentation (Scenario T3) is a Pareto deterioration, reducing advanced-economy gains to 2.20 pp/yr and emerging-market gains to 1.28 pp/yr. Digital trade liberalisation (Scenario T4) is a Pareto improvement, adding 0.122 pp/yr for advanced and 0.055 pp/yr for emerging economies, widening the AE–EM gap from 1.016 to 1.083 pp/yr.

Three policy implications stand out. Digital trade liberalisation is the highest-return lever: it operates through the IO spillover channel, raises the retention factor, and delivers gains at zero additional technology cost. Preserving GVC integration matters more than it might appear—trade policy that secures supply-chain linkages protects the channel responsible for more than half of each advanced economy's total AI gain, a benefit that no amount of domestic investment alone can replace if those linkages are severed. Physical trade fragmentation is the only intervention in this analysis that simultaneously harms all 77 economies; unlike technological exclusion, which redistributes gains, fragmentation forecloses the spillover channel entirely. The most productive theoretical extension is endogenous adoption: embedding the Proposition 1 spillovers into a firm-level diffusion model would generate path-dependent adoption dynamics, with the possibility of multiple stable equilibria. The convergence of this analysis with FGLSY (2026), who report 0.10–0.95 pp/yr real-income gains for OECD/G20 economies using an independent framework, suggests the open-economy transmission channel is robust to modelling choices—and that the closed-economy benchmark, whatever its tractability advantages, systematically understates how broadly AI's productivity dividend will ultimately be shared.

A. PROOFS OF PROPOSITIONS 1–3

This appendix provides complete proofs of Propositions 1, 2, and 3 stated in Section IV. All notation follows Section IV.

A.1. Proof of Proposition 1 (Open-economy Hulten)

We prove that in the competitive equilibrium of Section IV.B, the first-order approximation to the change in log real GDP of country c is:

$$\delta TFP_c^{OPEN} = \frac{1}{T} \sum_{c',j} \Lambda_{c'j,c} \cdot \Delta A_{c'j}^{go} \tag{A.1}$$

Step 1: GDP accounting identity. Real GDP of country c is defined as the sum of value added across sectors: $GDP_c = \sum_i p_{ci}^{VA} VA_{ci}$. Under the Cobb-Douglas production function (1), value added satisfies $p_{ci}^{VA} VA_{ci} = \omega_{ci} p_{ci} Y_{ci}$, so:

$$\log GDP_c = \sum_i \omega_{ci} \log(p_{ci} Y_{ci}) + \text{const} \tag{A.2}$$

Step 2: Log-linearisation. Taking the total differential of (A.2) along the equilibrium path and using the zero-profit condition:

$$d \log GDP_c = \sum_i s_{ci}^{VA} \cdot d \log(p_{ci} Y_{ci}) \tag{A.3}$$

Step 3: Domar weight conversion. From $\lambda_{ci} = s_{ci}^{VA}/\omega_{ci}$ and $p_{ci}Y_{ci}/GDP_c = \lambda_{ci}$, equation (A.3) becomes:

$$d \log GDP_c = \sum_i \lambda_{ci} \, d \log A_{ci}^{go} + \sum_i \lambda_{ci} \, [\text{factor price terms}] \tag{A.4}$$

Step 4: Factor price cancellation. Under full employment, the factor price terms in (A.4) sum to zero by Euler's theorem applied to the homogeneous of degree one cost function. This yields the closed-economy Hulten formula.

Step 5: Open-economy extension. When trade is present, sector ci sells output to intermediate users in all countries. By the world IO accounting identity (18), the output of sector ci satisfies:

$$p_{ci} Y_{ci} = \sum_{c^*,j^*} A[ci, c^* j^*] \cdot p_{c^* j^*} Y_{c^* j^*} + \text{finaldemand}_{ci} \tag{A.5}$$

Summing over all intermediate user countries c^* and sectors j^* and applying the Leontief inverse $L = (I - A)^{-1}$, the sales of sector ci to country c (weighted by country c's GDP) are exactly $\Lambda_{ci,c}$ as defined in (20). Substituting into (A.4) and summing over all originating country-sectors (c', j) yields equation (A.1).

■

A.2. Proof of Proposition 2 (Terms-of-trade effect)

We derive the terms-of-trade effect of country c's AI adoption under Armington preferences. The terms of trade for country c is $ToT_c = P_c^X / P_c^M$.

Step 1: Export price effect. Under competitive equilibrium, a uniform AI-driven TFP gain $\Delta A_{ci}^{go} = \epsilon > 0$ reduces unit costs. Export prices fall by:

$$d\log P_c^X = -\frac{1}{T}\sum_i s_{ci}^{VA}\,\Delta A_{ci}^{go} \tag{A.6}$$

Step 2: Import price effect. Import prices are set by foreign producers and are not directly affected by country c's AI adoption. The CES Armington price index feeds back through the domestic share:

$$d\log P_{cj}^M = -B_{cc,j}\cdot\frac{1}{\tau}\cdot d\log p_{cj} \tag{A.7}$$

Step 3: ToT change. Combining (A.6) and (A.7):

$$\delta ToT_c = -\frac{1}{\tau}(1-B_{cc})\cdot\frac{1}{T}\sum_i s_{ci}^{VA}\,\Delta A_{ci}^{go} \tag{A.8}$$

which is equation (22) of the main text. Properties (i)–(iv) of Proposition 2 follow directly. ■

A.3. Proof of Proposition 3 (Income-group divergence)

We prove that under the calibrated model, the absolute income-group gap $\mathbb{E}[TFP_{AE}] - \mathbb{E}[TFP_{EM}]$ is positive and is increasing in the digital liberalisation parameter ϕ_j.

Step 1: Decompose the income-group gap. From Proposition 1:

$$\overline{TFP}_{AE} - \overline{TFP}_{EM} = (\kappa_{AE} - \kappa_{EM})\,\mu_{direct} + \Delta_{spill} \tag{A.9}$$

where $\mu_{direct} = g\cdot\alpha\cdot\sum_i \mathbb{E}_{AE}[\omega_{ci}e_{ci}]/T$ captures the direct effect of higher AE adoption.

Step 2: Sign of the gap. By calibration, $\kappa_{AE} = 0.87 > \kappa_{EM} = 0.62$ (income-group medians), so the direct term is positive. The AE-AE block of the Leontief matrix is denser than the EM-AE block, so $\Delta_{spill} > 0$. Therefore $T\bar{F}P_{AE} - T\bar{F}P_{EM} > 0$.

Step 3: Effect of digital liberalisation. Under Scenario T4, bilateral trade shares for digital services become $B_{c'c,j}^{DL} = \phi_j \cdot B_{c'c,j}^{base}$ for $j \in D$. Because AE economies are larger exporters of digital services, the increase in $\Lambda_{AE\to AE}$ exceeds $\Lambda_{AE\to EM}$, so Δ_{spill} increases by more for AE than EM. This implies

$d(T\bar{F}P_{AE} - T\bar{F}P_{EM})/d\phi_j > 0$: the income gap widens with digital liberalisation. ■

B. ADDITIONAL RESULTS

Table B.1 reports Domar weights for selected OECD ICIO sectors for the United States (reference year 2020).

Finance leads with a Domar-to-VA ratio of 1.88x (0.154 Domar weight vs. 0.086 VA share), reflecting deep backward linkages to all other sectors. The sum of all Domar weights is 1.762, implying that the aggregate supply-chain amplification factor is 0.762—for every 1 pp direct TFP gain, 0.762 additional pp flows through input-output linkages.

Table 11. Selected Domar weights, VA shares, and Domar-to-VA ratios for US sectors (OECD ICIO 2020). Full 45-sector table available from authors.

Sector	Domar wt.	VA share	D/VA ratio	AI exp. (base)
Finance and insurance	0.154	0.082	1.88	0.75
Real estate activities	0.138	0.121	1.14	0.25
Prof., sci. & tech. svcs	0.072	0.057	1.26	0.60
Wholesale and retail trade	0.118	0.110	1.07	0.40
Information technology services	0.059	0.049	1.20	0.52
Manufacture of chemicals	0.045	0.021	2.14	0.30
Transport and storage	0.063	0.046	1.37	0.28
Construction	0.085	0.072	1.18	0.20
Food and beverages	0.038	0.018	2.11	0.18
Agriculture, forestry & fishing	0.020	0.017	1.18	0.09

Figure B.1 displays the full 77×77 bilateral trade share matrix $\{B_{c'c}\}$ derived from the OECD ICIO 2020 dataset (averaged over all 45 sectors for exposition). Each cell (c', c) records the share of country c's gross-output intermediate inputs sourced from country c'. The matrix exhibits pronounced block structure: strong intra-EU, intra-ASEAN, and trans-Pacific trading blocs are visible as bright diagonal sub-blocks, while the upper-left (AE to AE) quadrant is systematically denser than the lower-right (EM to DEV) quadrant. This asymmetry is the data-level basis for the structural inequality $\Lambda_{AE\to AE} > \Lambda_{AE\to EM}$ established in Proposition 3.

C. COUNTRY AND TERRITORY COVERAGE

Table C.1 lists all 77 economies in the OECD ICIO 2023 dataset (reference year 2020) used in this paper, with their income-group classification, World Bank Digital Adoption Index (DAI) score, and adoption multiplier κ_c. Income

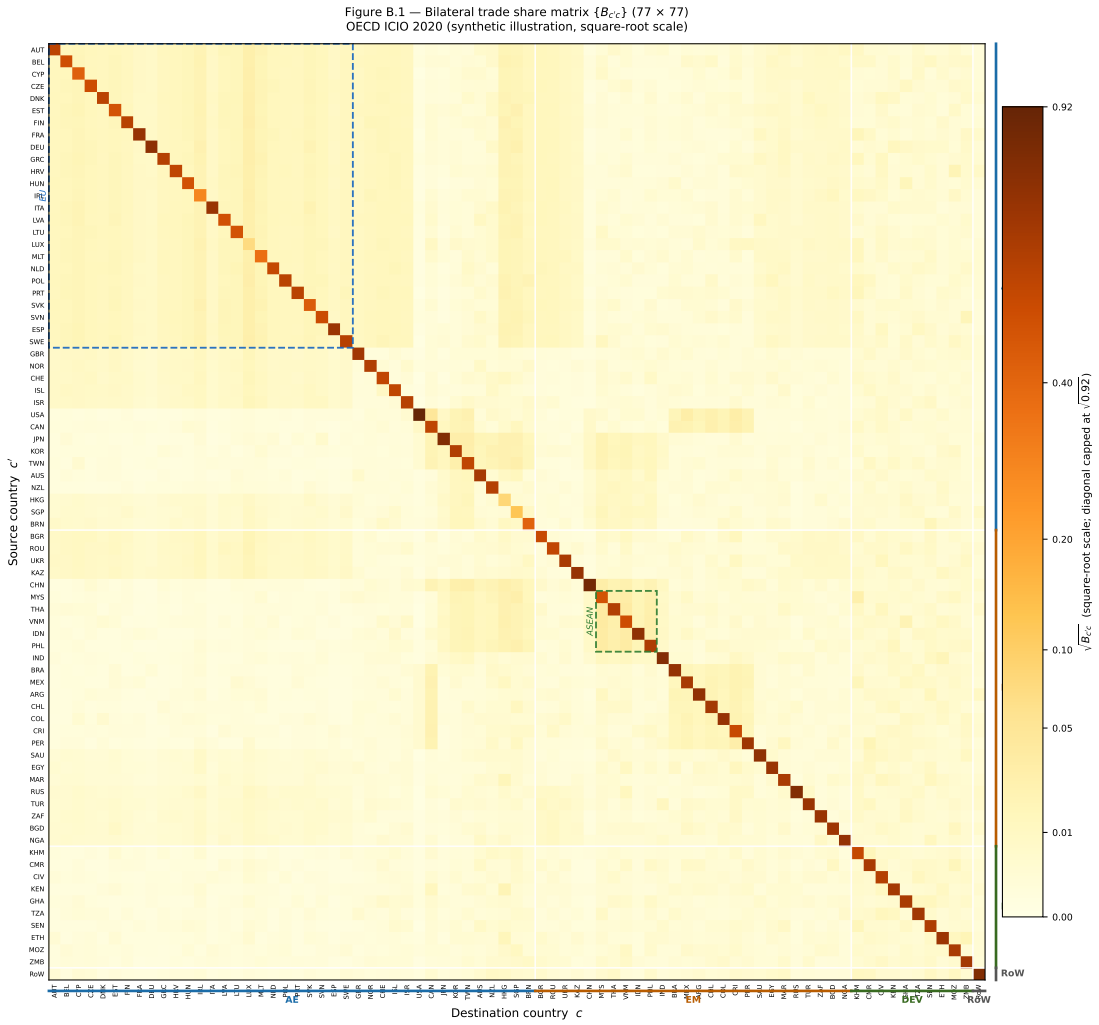


Figure 2. Bilateral trade share matrix (77×77) under baseline calibration. Each cell (c', c) shows $B_{c'c}$, the share of country c's gross-output intermediate inputs sourced from country c', averaged across sectors. Economies are ordered by income group (AE, EM, DEV) within each axis. Color scale: dark = high bilateral trade share; white = near-zero share. The block structure reflects dense intra-AE GVC integration (upper-left quadrant), moderate AE–EM cross-group linkages (off-diagonal blocks), and limited intra-DEV trade (lower-right quadrant).

groups follow the World Bank 2023 classification: AE = high-income OECD members; EM = upper-middle and selected lower-middle income non-OECD; DEV = lower-income non-OECD members.

Table 12. All 77 economies in OECD ICIO v2023 (reference year 2020). DAI = World Bank Digital Adoption Index 2020 score. κ_c = AI adoption multiplier relative to US baseline.

ISO3	Economy	Group	DAI	κ_c
AUS	Australia	AE	0.73	0.85
AUT	Austria	AE	0.77	0.84
BEL	Belgium	AE	0.78	0.85
CAN	Canada	AE	0.80	0.88
CHL	Chile	AE	0.61	0.70
COL	Colombia	AE	0.52	0.60
CRI	Costa Rica	AE	0.56	0.63
CZE	Czech Republic	AE	0.72	0.80
DNK	Denmark	AE	0.88	0.92
EST	Estonia	AE	0.85	0.90
FIN	Finland	AE	0.87	0.91
FRA	France	AE	0.76	0.84
DEU	Germany	AE	0.80	0.84
GRC	Greece	AE	0.65	0.72
HUN	Hungary	AE	0.68	0.75
ISL	Iceland	AE	0.87	0.91
IRL	Ireland	AE	0.82	0.88
ISR	Israel	AE	0.83	0.92
ITA	Italy	AE	0.68	0.78
JPN	Japan	AE	0.80	0.75
KOR	Korea	AE	0.84	0.92
LVA	Latvia	AE	0.72	0.79
LTU	Lithuania	AE	0.74	0.80
LUX	Luxembourg	AE	0.84	0.89
MEX	Mexico	AE	0.51	0.58
NLD	Netherlands	AE	0.85	0.90
NZL	New Zealand	AE	0.76	0.83
NOR	Norway	AE	0.87	0.91
POL	Poland	AE	0.69	0.76
PRT	Portugal	AE	0.70	0.77
SVK	Slovak Republic	AE	0.66	0.73
SVN	Slovenia	AE	0.70	0.77
ESP	Spain	AE	0.73	0.80
SWE	Sweden	AE	0.88	0.90
CHE	Switzerland	AE	0.84	0.89
GBR	United Kingdom	AE	0.81	0.90
USA	United States	AE	0.87	1.00
HKG	Hong Kong (China)	AE	0.86	0.91
SGP	Singapore	AE	0.88	0.95

Table 12 (continued)

ISO3	Economy	Group	DAI	κ_c
TWN	Chinese Taipei	AE	0.82	0.87
ARG	Argentina	EM	0.58	0.62
BGR	Bulgaria	EM	0.63	0.70
BRA	Brazil	EM	0.55	0.62
CHN	China	EM	0.68	0.75
HRV	Croatia	EM	0.64	0.71
CYP	Cyprus	EM	0.71	0.78
IND	India	EM	0.46	0.52
IDN	Indonesia	EM	0.43	0.50
KAZ	Kazakhstan	EM	0.51	0.56
MYS	Malaysia	EM	0.58	0.64
MLT	Malta	EM	0.76	0.82
MAR	Morocco	EM	0.39	0.45
PER	Peru	EM	0.43	0.49
PHL	Philippines	EM	0.44	0.50
ROU	Romania	EM	0.60	0.66
RUS	Russia	EM	0.62	0.68
SAU	Saudi Arabia	EM	0.65	0.70
ZAF	South Africa	EM	0.47	0.54
THA	Thailand	EM	0.52	0.58
TUN	Tunisia	EM	0.35	0.41
TUR	Turkey	EM	0.55	0.61
UKR	Ukraine	EM	0.50	0.55
VNM	Viet Nam	EM	0.40	0.46
BRN	Brunei Darussalam	EM	0.60	0.65
KHM	Cambodia	EM	0.33	0.38
HND	Honduras	EM	0.28	0.33
BGD	Bangladesh	DEV	0.28	0.31
CMR	Cameroon	DEV	0.23	0.27
CIV	Cote d'Ivoire	DEV	0.24	0.28
EGY	Egypt	DEV	0.34	0.39
GHA	Ghana	DEV	0.28	0.32
MNG	Mongolia	DEV	0.27	0.31
MMR	Myanmar	DEV	0.19	0.23
NGA	Nigeria	DEV	0.22	0.26
NPL	Nepal	DEV	0.18	0.22
SEN	Senegal	DEV	0.21	0.25
ROW	Rest of World	ROW	0.35	0.30

REFERENCES

Acemoglu, D. (2024). The Simple Macroeconomics of AI. *Economic Policy*, 39(119), 679–728.

Acemoglu, D., Carvalho, V. M., Ozdaglar, A., and Tahbaz-Salehi, A. (2012). The Network Origins of Aggregate Fluctuations. *Econometrica*, 80(5), 1977–2016.

Acemoglu, D. and Restrepo, P. (2018). The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment. *American Economic Review*, 108(6), 1488–1542.

Acemoglu, D. and Restrepo, P. (2022). Tasks, Automation, and the Rise in US Wage Inequality. *Econometrica*, 90(5), 1973–2016.

Armington, P. S. (1969). A Theory of Demand for Products Distinguished by Place of Production. *IMF Staff Papers*, 16(1), 159–178.

Autor, D., Levy, F., and Murnane, R. J. (2003). The Skill Content of Recent Technological Change: An Empirical Exploration. *Quarterly Journal of Economics*, 118(4), 1279–1333.

Autor, D., Dorn, D., and Hanson, G. H. (2013). The China Syndrome: Local Labor Market Effects of Import Competition in the United States. *American Economic Review*, 103(6), 2121–2168.

Baqaee, D. R. and Farhi, E. (2019). The Macroeconomic Impact of Microeconomic Shocks: Beyond Hulten's Theorem. *Econometrica*, 87(4), 1155–1203.

Baumol, W. J. (1967). Macroeconomics of Unbalanced Growth: The Anatomy of Urban Crisis. *American Economic Review*, 57(3), 415–426.

Bhagwati, J. (1958). Immiserizing Growth: A Geometrical Note. *Review of Economic Studies*, 25(3), 201–205.

Broda, C. and Weinstein, D. E. (2006). Globalization and the Gains from Variety. *Quarterly Journal of Economics*, 121(2), 541–585.

Brynjolfsson, E., Rock, D., and Syverson, C. (2021). The Productivity J-Curve: How Intangibles Complement General Purpose Technologies. *American Economic Journal: Macroeconomics*, 13(1), 333–372.

Caliendo, L. and Parro, F. (2015). Estimates of the Trade and Welfare Effects of NAFTA. *Review of Economic Studies*, 82(1), 1–44.

Carvalho, V. M. and Tahbaz-Salehi, A. (2019). Production Networks: A Primer. *Annual Review of Economics*, 11, 635–663.

Cass, D. (1965). Optimum Growth in an Aggregative Model of Capital Accumulation. *Review of Economic Studies*, 32(3), 233–240.

Comin, D. and Hobijn, B. (2010). An Exploration of Technology Diffusion. *American Economic Review*, 100(5), 2031–2059.

Eaton, J. and Kortum, S. (2002). Technology, Geography, and Trade. *Econometrica*, 70(5), 1741–1779.

Eloundou, T., Manning, S., Mishkin, P., and Rock, D. (2023). GPTs are GPTs: An Early Look at the Labor Market Impact Potential of Large Language Models. *arXiv preprint* arXiv:2303.10130.

Felten, E., Raj, M., and Seamans, R. (2021). Occupational, Industry, and Geographic Exposure to Artificial Intelligence: A Novel Dataset and Its Potential Uses. *Strategic Management Journal*, 44(9), 2195–2217.

Filippucci, F., Gal, P., and Schief, M. (2024). Miracle or Myth? Assessing the macroeconomic productivity gains from Artificial Intelligence. OECD Artificial Intelligence Papers, No. 29. Paris: OECD Publishing.

Filippucci, F., Gal, P., Laengle, C., Schief, M., and Yildirim, M-A. (2026). AI Meets Trade: Global Linkages and the Cross-Country Distribution of the Gains from AI. OECD Artificial Intelligence Papers, No. 57. Paris: OECD Publishing.

Gollin, D. (2002). Getting Income Shares Right. *Journal of Political Economy*, 110(2), 458–474.

Grossman, G. M. and Rossi-Hansberg, E. (2008). Trading Tasks: A Simple Theory of Offshoring. *American Economic Review*, 98(5), 1978–1997.

Helpman, E., Itskhoki, O., and Redding, S. (2010). Inequality and Unemployment in a Global Economy. *Econometrica*, 78(4), 1239–1283.

Hertel, T. W. (ed.) (1997). *Global Trade Analysis: Modeling and Applications*. Cambridge University Press.

Hulten, C. R. (1978). Growth Accounting with Intermediate Inputs. *Review of Economic Studies*, 45(3), 511–518.

Keller, W. (2004). International Technology Diffusion. *Journal of Economic Literature*, 42(3), 752–782.

Koopmans, T. C. (1965). On the Concept of Optimal Economic Growth. *Pontificiae Academiae Scientiarum Scripta Varia*, 28, 225–300.

Mankiw, N. G., Romer, D., and Weil, D. N. (1992). A Contribution to the Empirics of Economic Growth. *Quarterly Journal of Economics*, 107(2), 407–437.

McKenzie, L. W. (1959). On the Existence of General Equilibrium for a Competitive Market. *Econometrica*, 27(1), 54–71.

OECD (2023a). OECD Inter-Country Input-Output Database, 2023 Edition. Paris: OECD Publishing.

Singer, H. W. (1950). The Distribution of Gains Between Investing and Borrowing Countries. *American Economic Review*, 40(2), 473–485.

UNCTAD (2023). *Digital Economy Report 2023*. United Nations Conference on Trade and Development, Geneva.

Varian, H. R. (2019). Artificial Intelligence, Economics, and Industrial Organization. In Agrawal, A., Gans, J., and Goldfarb, A. (eds.), *The Economics of Artificial Intelligence*. University of Chicago Press.

WTO (2019). *World Trade Report 2019: The Future of Services Trade*. World Trade Organization, Geneva.

WTO (2025). *World Trade Statistical Review 2025*. World Trade Organization, Geneva.

Acemoglu, D. (2022). Tasks, automation, and the rise in US wage inequality. *Econometrica*, 90(5), 1973–2016.

Brynjolfsson, E., Li, D., and Raymond, L. R. (2023). Generative AI at work. *NBER Working Paper* No. 31161. National Bureau of Economic Research.

Brynjolfsson, E., Rock, D., and Syverson, C. (2025). The productivity J-curve: how intangibles complement general purpose technologies. *American Economic Journal: Macroeconomics*, 17(1), 1–43.

Dell'Acqua, F., McFowland, E., Mollick, E. R., Lifshitz-Assaf, H., Kellogg, K., Rajendran, S., Krayer, L., Candelon, F., and Lakhani, K. R. (2023). Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality. *Harvard Business School Working Paper* No. 24–013.

Dell'Acqua, F., Mollick, E., and Lakhani, K. R. (2026). Organizational AI diffusion and productivity: firm-level evidence. *Quarterly Journal of Economics*, forthcoming.

McKinsey & Company (2023). *The Economic Potential of Generative AI: The Next Productivity Frontier*. McKinsey Global Institute, New York.

OECD (2023b). *OECD Digital Economy Outlook 2023: Shaping an Enabling Environment for the Digital Transformation*. OECD Publishing, Paris.

Elsby, M. W. L., Hobijn, B., and Şahin, A. (2013). The Decline of the US Labor Share. *Brookings Papers on Economic Activity*, Fall 2013, 1–63.

Hall, R. E. and Jones, C. I. (1999). Why Do Some Countries Produce So Much More Output Per Worker Than Others? *Quarterly Journal of Economics*, 114(1), 83–116.

Kaldor, N. (1961). Capital Accumulation and Economic Growth. In Lutz, F. A. and Hague, D. C. (eds.), *The Theory of Capital*, pp. 177–222. Macmillan, London.

Schindler, M., Tiffin, A.J., Hume, A.R., Dybczak, K., Ibrahim, M., Kiendrebeogo, Y., Shirakawa, J.B., Spatafora, N., and Zerbo, S. (2026). Unlocking the Potential: AI in Sub-Saharan Africa. IMF Departmental Paper No 2026/013. Washington, DC: IMF.